

Mixed bus fleet scheduling under range and refueling constraints

Lu Li^{a,b}, Hong K. Lo^{c,*}, Feng Xiao^d

^a Business School, Sichuan University, 610065 Chengdu, PR China

^b California PATH, University of California, Berkeley, Richmond, CA, United States

^c Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Hong Kong, China

^d School of Business Administration, Southwestern University of Finance and Economics, PR China

ARTICLE INFO

Keywords:

Bus scheduling
Mixed fleet
Electric bus
Driving range
Refueling/charging

ABSTRACT

This paper develops a formulation for the multiple depot (MD) vehicle scheduling problem with multiple vehicle types (MVT), including electric buses (EBs), under range and refueling constraints. A novel approach is developed to generate the feasible time-space-energy (TSE) network for bus flow and time-space (TS) network for passenger flow, where the range and refueling issues can be precisely addressed. We then introduce the external cost associated with emissions, and investigate the minimum total system cost to operators and passengers by scheduling the bus fleet and locating the refueling stations. The problem is formulated as an integer linear program (ILP) to find the global optimal solution. For computational efficiency, we develop a simplified formulation based on the TS bus flow network to handle larger-scale problems for approximate solutions. We apply the methods to bus services in Hong Kong to analyze the bus fleet size needed, operational cost, passenger cost, and emissions generated by buses with multiple energy sources. Through the formulation, we study the implications of government subsidy for EBs, bus scheduling in Low Emission Zone (LEZ), and safe driving ratio design for bus refueling.

1. Introduction

Environmental sustainability has recently emerged as an important factor driving public policy. The European Commission has reported that poor air quality causes more premature deaths than road accidents, responsible for 0.31 million premature deaths in Europe every year. In public transportation, majority of buses are primarily heavy-duty diesel vehicles, which produce a substantial amount of air pollution (US EPA, 2008). Emissions from buses are particularly severe for transit-oriented metropolitans, such as Hong Kong, where buses carry the majority of trips. To improve roadside air quality, over 220 cities and towns in 14 countries around Europe are operating or preparing to establish Low Emission Zones (LEZs). This has led to subsidy policies to promote alternative energy vehicles, such as hybrid and electric buses (EBs), to reduce or remove tail-pipe emissions. During the process, the range concern arising from electric or alternative energy buses becomes an important issue due to the high daily mileages of buses. In any case, in the deployment of alternative energy buses, particularly EBs, range constraints must be duly incorporated in bus route assignment and scheduling.

The traditional bus-scheduling problem is to cover all the trips in the timetable with fixed travel times and start and end locations. The objectives are to minimize the bus fleet size and operating cost. Bunte and Kliwer (2009) gave an overview on vehicle scheduling problems (VSP) and discussed several modeling approaches. Earlier bus-scheduling studies seldom considered energy and emissions, whose main idea was to minimize the fleet size. Conventional bus service, which has fixed service schedules operated by a

* Corresponding author.

E-mail address: cehklo@ust.hk (H.K. Lo).

<https://doi.org/10.1016/j.trc.2019.05.009>

Received 22 March 2018; Received in revised form 28 March 2019; Accepted 9 May 2019

Available online 29 May 2019

0968-090X/ © 2019 Elsevier Ltd. All rights reserved.

single bus type, is not always cost-effective due to variable demand densities. Hence, some studies were conducted to investigate the multi-vehicle-type bus scheduling problem (MVT-VSP), involving different vehicle types of diverse capacities for timetabled trips (Hassold and Ceder, 2012, 2014; Ceder et al., 2013; Kim and Schonfeld, 2014). It is worth noting that the majority of research on VSP is trip-based. Only a few considered time-dependent passenger demand by using passenger waiting time to reschedule the service trips. Hassold and Ceder (2012) and Ceder et al. (2013) investigated the problem of how to make public bus services more attractive. Both of them aimed to minimize the passenger waiting time to improve public transport reliability. Lai and Lo (2004), Wang and Lo (2008), Lo et al. (2013), An and Lo (2014, 2015, 2016) considered the passenger cost when they formulated the network design problem on transit and ferry services. In addition to fulfilling all the timetabled trips, this study seeks to find a most cost-effective design to balance the costs between the operator and passengers via the waiting penalty, which will improve both the utilization of buses and attractiveness of the services.

Besides, it is inefficient to restrict each bus to operate on a single route. Kliewer et al. (2006) developed time-space networks for MVT-VSP to allow vehicles to cover another service route after finishing one timetabled trip, which was called bus service coordination. Kliewer et al., (2008) extended the study by involving several line changes for a given bus within a working day. This mixed-line scheduling scheme can make fuller use of the bus fleet in the network, thus reducing both the operational cost and bus fleet needed. In this paper, we adopt a similar approach and incorporate mixed-fleet mixed-route scheduling to take full utilization of the bus fleet.

In recent years, more and more VSP studies considered clean-energy buses due to environmental and energy concerns. Due to range constraints and long recharging time, the approach to schedule the bus fleet is quite different from the conventional VSP. Li (2013) proposed a vehicle scheduling model for EBs, as well as for compressed natural gas (CNG), diesel, and hybrid buses, respectively, with maximum route distance constraints. Zhu and Chen (2013) established a single depot vehicle scheduling model for EBs, in which the charging time has a linear relationship with the corresponding bus mileage. Both of these studies considered a single fleet VSP and set the timetable for bus charging according to its maximum driving distance. Notice that using travel distance to determine the driving range is a rough proxy, since energy consumption is not only determined by distance, but also by travel speed, passenger loading, gradient of the terrain, battery temperature, and current battery life (Thein and Chang, 2014; Goeke and Schneider, 2015), etc. In this paper, we incorporate energy consumption into formulating the bus flow network by considering the impacts of travel speed and passenger loading.

Locating refueling stations is another indispensable issue for incorporating range constraints in VSP. He et al. (2013) proposed a macroscopic planning model to design the optimal number of charging stations allocated to each metropolitan area without considering the exact locations and capacities. Then they proceeded to optimize the deployment plan of charging lanes for electric vehicles over a general network (Chen et al., 2016). Lim and Kuby (2010) developed three heuristic algorithms to locate the refueling stations for alternative-fuels using path-based demands. Schneider et al. (2014) presented an electric vehicle routing problem, which considered setting a set of available recharging stations beforehand. Besides, some novel recharging facilities were also investigated by recent studies. Liu and Wang (2017) proposed a modeling framework for locating multiple types of battery electric vehicle charging facilities, who considered wireless static and dynamic charging in the decision procedure. In this paper, we generate the bus flow network by considering a set of feasible candidate refueling stations to tackle the charging station locating problem.

For the mixed fleet VSP concerning alternative energy sources, Li and Head (2009) established a bus-scheduling model to minimize the operating cost and excess vehicle emissions involving CNG and Hybrid buses, while range constraints and charging time were not taken into account. Beltran et al. (2009) and Pternea et al. (2015) focused on developing an efficient model to solve a sustainability-oriented variant of the transit route network design problem. Both of them assigned electric vehicles to pre-determined routes without considering their charging time. By using a set of given weights, the objective functions sought to minimize the user, operator and external costs. A model of transit design for a mixed bus fleet was developed by Fusco et al. (2013) to compute the internal and external costs during the life cycle of a bus. It introduced a fleet of EBs to operate some lines of a transit corridor in an urban area, in which charging facilities were considered whereas the assignment of travel routes for each electric bus was not. Goeke and Schneider (2015) optimized the routes of a mixed fleet of electric and diesel commercial vehicles to fulfill the customer demand. A routing model was developed to design the routes for each electric vehicle. Yet the model was not built for VSP, and emission was not taken into consideration. Actually, few studies have been conducted on the mixed bus fleet scheduling problem with regards to range and refueling constraints, and emissions reduction simultaneously. This paper considers not only the daily travel routes of two types of buses with different energy sources, but also different bus capacities to handle the passenger demand.

In our previous work, we considered the mixed bus fleet management and government subsidy scheme for early-retiring, purchasing, and routing problems based on the frequency-based approach (Li et al., 2015, 2018). In this paper, we consider the problem based on the schedule-based approach, which can explicitly deal with the range and refueling issues, as well as some additional problems, such as locating the refueling stations, considering passengers route choices, etc. In most previous studies on electric bus scheduling, they used time-space (TS) network to develop non-linear programs to solve for approximate solutions. Since energy is not tracked in these studies, range and refueling (charging time) constraints cannot be formulated linearly. In this study, we incorporate the energy consumption state variable explicitly into our model, and develop a novel approach to generate a time-space-energy (TSE) network for bus flow which linearly and precisely addresses the range and refueling constraints in VSP. Besides, we also build a TS network for passenger flow which allows for detour travels. Based on these two feasible networks, we develop an integer linear program (ILP) to formulate the multiple depot (MD) VSP with multiple vehicle types (MVT) to achieve global optimality, referred to here as MD-MVT-VSP. We minimize the costs of both operators and passengers, as well as the external cost caused by emissions, by assigning schedules and trips to a mixed fleet of EBs and diesel buses (DBs). We also incorporate the problem of locating refueling stations into our VSP and optimize them simultaneously. To speed up the computational process, we develop a simplified formulation

to solve for larger-scale problems with approximate solutions. The results are promising; in the numerical case study, the number of variables is reduced by 96%, and computational time by 83%. It is worth noting that the global optimum can also be obtained by using the latter approach when (1) the planning horizon is within twice the travel time range of a fully refueled bus; and (2) the time interval of the network is larger than or equal to the time needed to fully refuel a bus. Compared with the conventional VSP, the proposed MD-MVT-VSP has distinctive advantages. The main contributions of this paper are as follows:

- (1) We consider the mixed bus fleet scheduling problem with bus service coordination. By integrating bus service, passenger movement, and bus emissions into one model, the optimal bus scheduling scheme is determined;
- (2) We propose a novel approach to generate the feasible TSE network for bus flow by capturing energy consumption explicitly. The range and refueling issues are precisely tackled based on the TSE networks generated, as is the refueling station location problem;
- (3) We formulate the VSP as an ILP to find the global optimal solution instead of relying on heuristics as adopted by most studies. Furthermore, for computational efficiency, we develop alternative formulations based on the TS networks to handle larger-scale problems.

The outline of this paper is as follows: [Section 2](#) describes the problem. [Section 3](#) presents the methodology to solve the MD-MVT-VSP based on the TSE network approach and proposes an alternative approach to obtain approximate solutions with reduced computational time. [Section 4](#) examines the performances of the two approaches developed in [Section 3](#) by applying it to a small network and then to an actual transit network in Hong Kong; and [Section 5](#) provides some concluding remarks. For ease of referencing, we provide the notation in [Appendix A](#) and acronyms in [Appendix B](#).

2. Problem description

The multiple depot vehicle scheduling problem with multiple vehicle types (MD-MVT-VSP) in this paper can be stated as follows: For a given set of depots and transit routes, given travelling distances between all pairs of bus termini, find the minimum system cost by determining the bus routing and service schedule within the planning horizon. We only need to consider the starting and ending termini in the formulation without involving the intermediate stops, as each bus will finish a complete trip and will not switch to another trip midway. The system cost involves the cost to operators and passengers, as well as the external cost associated with emissions. Unlike the traditional bus-scheduling problem, passenger-waiting and demand-loss costs are introduced so that not all demand has to be satisfied. MD-MVT-VSP seeks to balance the benefits between the operator and the passengers from the social welfare perspective.

A vehicle schedule is feasible if (1) each trip of the scheduled timetable is covered exactly once, (2) each vehicle performs a feasible sequence of trips where each pair of consecutive trips can be operated in sequence, (3) each vehicle has sufficient energy to travel the next trip, (4) each vehicle is put into use again when it finishes its refueling process and (5) each vehicle returns to its starting depot at the end of its daily travel route. MVT-VSP considers a heterogeneous fleet of vehicles using alternative energies. Since each vehicle type has its own driving range, its energy consumption and energy capacity are recorded for each trip for determining the specific moment at which refueling has to take place at a refueling station to avoid energy shortage.

For simplicity, we make the following assumptions:

- (1) Each transit route consists of bi-directional transit lines between two termini. By setting two bus termini instead of one with zero distance between them, the proposed model is also applicable for circular lines.
- (2) The demand for each origin-destination (OD) pair is given, and its arrival pattern at each terminus follows a trip distribution which is known beforehand.
- (3) The refueling time of each vehicle type follows a refueling rule, either linear or non-linear, with respect to the cumulative energy consumption, and full energy is restored after each refueling.

3. Modeling approach

In this section, we first propose an approach to solve the multiple depot vehicle scheduling problem with multiple vehicle types based on the TSE network, referred to as MD-MVT-VSP-TSE in this study. Since energy consumption is tracked as a state variable or label in the network, the range and refueling issues can be precisely addressed. As an ILP, global optimality can be achieved through the formulation. To reduce the number of variables and speed up the computational process, we develop an alternative approach based on the TS network to solve the MD-MVT-VSP, referred to as MD-MVT-VSP-TS without tracking energy consumption explicitly. This approach is proposed to handle larger-scale networks with approximate solutions, where global optimal solutions can be obtained if the refueling times of buses under range constraints are smaller than the duration of the time interval in the network, as will be explained later.

3.1. MD-MVT-VSP based on the TSE network (MD-MVT-VSP-TSE)

We first introduce the notation to describe the TSE network which is the basis of the mathematical formulation.

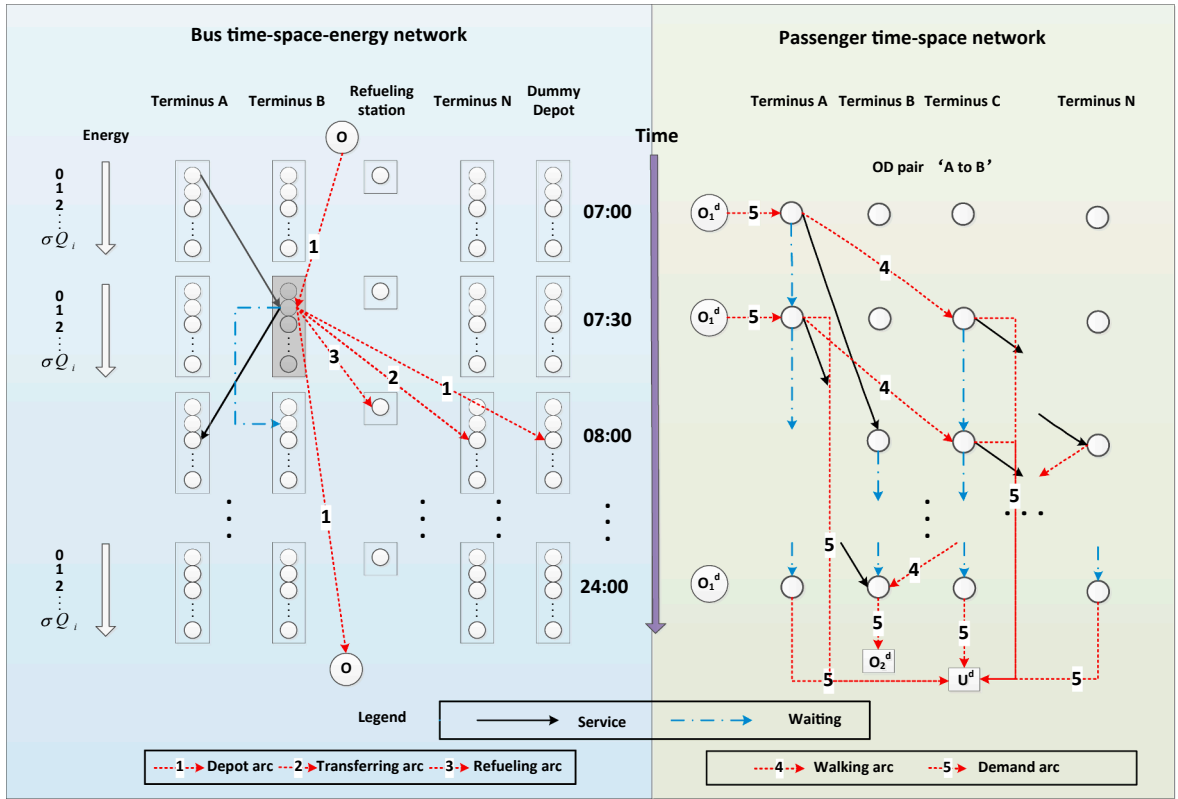


Fig. 1. The bus time-space-energy network (left) and passenger time-space network (right).

3.1.1.1. Generation of the feasible network

3.1.1.1.1. Bus TSE network. We generate the feasible daily network for each vehicle type with a specific energy source to keep track of each bus's energy consumption. For each depot an independent network is built. In the TSE network of bus flow, each node represents a specific location at a specific time with a specific level of energy consumption. Time is divided into a set of intervals by a fixed time duration, and energy consumption is divided into a set of levels by an energy step. In other words, at each time interval of each bus terminus, there are a set of nodes demarcated by different energy consumption levels. By keeping track of the energy consumption at each node, the remaining driving range can be calculated. A network is then built by connecting the arcs of all possible routes of the concerned bus type for each depot. In this way, not only do such TSE bus networks ensure that all buses travel within their energy limits, revealing refueling needs, they also allow for the incorporation of mixed-route bus trips.

Let $I = \{1, 2, \dots, i\}$ be the set of bus types with different energy sources, and $O = \{1, 2, \dots, o\}$ be the set of depots. The TSE network is defined by a graph $G(V, A_{io})$ as shown in Fig. 1. V represents the set of nodes with $V = O \cup T \cup F \cup O'$, where T denotes the set of time-energy-expanded bus termini, and F the set of time-expanded refueling stations, and O' denotes the set of dummy nodes for each depot indicating the temporary stay of buses at depot. A_{io} is the set of arcs with three subsets: service arc set S_{io} , waiting arc set W_{io} , and deadheading arc (the movement of a vehicle to a destination without serving any passenger) set D_{io} , such that $A_{io} = S_{io} \cup W_{io} \cup D_{io}$. Each service arc indicates a direct service trip between an OD pair starting at a specific time with a specific level of energy consumption, whereas each waiting arc connects two consecutive time nodes of the same terminus while maintaining the same energy level. Each deadheading arc represents a trip (i) from or to a depot, or (ii) to the starting terminus of another service trip, or (iii) to a refueling station.

Two parameters are introduced to adjust the travel time and energy consumption. One is γ_v^{jk} , the adjustment parameter for speed related to the time of day when traveling on the arc (j, k) . The other is γ_m^{jk} , the adjustment parameter for vehicle mass related to the passenger loading or demand. The travel time and energy consumption on each arc are determined by the travel distance d^{jk} , the average travel speed v , and the energy consumption rate η_i alongside with the two adjustment parameters. We define energy consumption as the cost of the arc (j, k) , i.e., $q_i^{jk} = d^{jk} \eta_i \gamma_v^{jk} \gamma_m^{jk}$, whereas travel time is denoted by $t^{jk} = \frac{d^{jk}}{v \gamma_v^{jk}}$. For bus type i , let Q_i be the energy capacity, τ_i^j be the time of arrival at node j , and e_i^j the cumulative energy consumption upon arrival at node j . Finally, let u_{io}^{jk} be an indicator specifying whether arc (j, k) is connected in the network $G(V, A_{io})$. The TSE network $G(V, A_{io})$ of bus type i can be represented by the following equations:

$$u_{io}^{ok} = 1 \quad \forall k \in \{k | k \in T, e_i^k = q_i^{ok}, e_i^k \leq \sigma Q_i\} \quad (1)$$

$$u_{io}^{j0} = 1 \quad \forall j \in \{j | j \in T \cup O', e_i^j > 0\}, o \in \{o | o \in O, \tau_i^o = \tau_i^j + t^{j0}, e_i^o = e_i^j + q_i^{j0}, e_i^o \leq \sigma Q_i\} \quad (2)$$

$$u_{io}^{jk} = 1 \quad \forall j \in T \cup F_i \cup O', k \in \{k | k \in T \cup O', \tau_i^k = \tau_i^j + t^{jk}, e_i^k = e_i^j + q_i^{jk}, e_i^k \leq \sigma Q_i\} \quad (3)$$

$$u_{io}^{jf} = 1 \quad \forall j \in \{j | j \in T, \sigma' Q_i \leq e_i^j\}, f \in \{f | f \in F_i, \tau_i^f = \tau_i^j + t^{jf}, e_i^f = e_i^j + q_i^{jf}, e_i^f \leq \sigma Q_i, e_i^f = 0\} \quad (4)$$

Define the fixed time step and energy step to be ξ_t and ξ_e . We first calculate the exact arrival time τ_i^k and cumulative energy consumption e_i^k of each node, and put it into the corresponding time interval and energy levels in the network, i.e. $\lceil \tau_i^k / \xi_t \rceil$ and $\lceil e_i^k / \xi_e \rceil$.

Constraints (1) and (2) specify the deadheading arcs from or to the depots. Let σ be the reduction factor of the energy capacity for planning purposes, namely safe driving ratio, to avoid energy depletion of the bus in operation, say 80 or 90%. Constraint (3) defines the connectivity of trips to bus termini and dummy depots, and the arrival time and cumulative energy consumption of the destination node are computed. Meanwhile, it guarantees that the maximum energy capacity cannot be violated with a safe driving ratio σ to avoid stranding. Let F_i be the refueling stations for buses of type i . All buses with low energy levels should still have enough energy to travel to a refueling station. To ensure this, we introduce a start refueling ratio σ' (say, 10%), so buses with energy levels at or just above $\sigma' Q_i$ should initiate refueling, as shown in constraint (4), who handles the connectivity of visits to a refueling station. Let t_i^r be the refueling time for buses of type i , a pre-determined parameter, which can also be modeled as a function of the energy capacity, either linear or non-linear, or a fixed value if we use the charging technology of battery swapping. Constraint (4) also defines the refueling process, and resets the cumulative energy consumption to be zero. Notice that there is only one energy level at each time interval at each refueling station, i.e. $e_i^f = 0$. To further reduce the network size, we can add $\sum_{p \in V} u_{io}^{pj} > 0$ in (2)–(4) for node j to avoid redundant connections if j is an isolated node without inflows.

Fig. 1 (left) illustrates the TSE network of bus type i . Time is divided by a fixed duration of 30 mins. Each rectangle consists of a set of nodes associated with the energy consumption levels at the same time and the same location, except for the ones in the refueling stations which only have one node with zero energy consumption. Take the gray rectangle at terminus B at 07:30 as an example; it is composed of five energy levels if we divide the energy consumption by an energy step of $\sigma Q_i / 4$. Specifically, take the node in the second energy level to illustrate, on one hand, it has two input links, i.e., (1) depot arc from depot where one energy level is consumed; and (2) service arc from terminus A indicates that the time duration of this trip is within 30 mins, which is 1 time interval, as is the energy level, which is within 1 energy step. On the other hand, it has six output links, i.e., (1) service arc to terminus A where this arc flows into the third node at the third time interval at terminus A; (2) waiting arc to the node with the same energy level after 30 min; (3) depot arc to depot indicating the completion of the whole day trip; (4) depot arc to dummy depot indicating the temporary return of buses to the depot; (5) transferring arc to the starting terminus of another service trip; and (6) refueling arc to the refueling station. To further explain the notation in (1)–(4), we display a small bus time-space-energy network with four labelled nodes that each indicates a specific location, time, and energy level, as shown in Fig. 2.

3.1.1.2. Passenger TS network. The TS network of passenger flow is defined by a set of graphs $G(V^d, A^d)$, where d refers to a particular OD pair belonged to the set of OD pairs R . Let V^d be the set of nodes with $V^d = O^d \cup T^d \cup U^d$ where O^d denotes the set of time-expanded passenger demands composed of original demands at the beginning of the daily service, O_1^d , and served demands at the end of the daily service, O_2^d , and U^d the set of unserved demands during the daily service. T^d is the set of time-expanded bus termini, where T_1^d is the time-expanded departure set and T_2^d is the time-expanded destination set with respect to OD pair d . Let A^d be the set of arcs with three subsets: service arc set S^d , waiting arc set W^d , and detouring arc set D^d , i.e. $A^d = S^d \cup W^d \cup D^d$. In the detouring arc set, there are two types of sub-arc sets; one is walking arc set D_1^d , and the other is demand arc set D_2^d such that $D^d = D_1^d \cup D_2^d$. Note that in each graph $G(V^d, A^d)$, there are three types of demand arcs: origin demand arc from nodes in O_1^d to nodes in T_1^d at the corresponding time points, served demand arc from the last node in T_2^d to node in O_2^d , and lost demand arc from nodes in $T^d \setminus T_2^d$ to U^d . Each service arc denotes a trip at a specific time, whose arc cost is the passenger traveling time cost. The flow on service arc denotes the number of onboard passengers, subject to the capacities of the bus services. The flow on the waiting arc, however, indicates the accumulated passengers not getting served and have to wait for the subsequent bus due to insufficient capacity of the departed bus. Walking arc describes the movement of passengers between locations within walking distance. The origin demand arc carries the

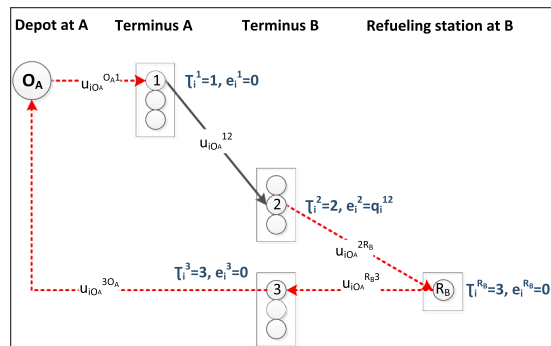


Fig. 2. Illustration of the bus time-space-energy network for bus type i .

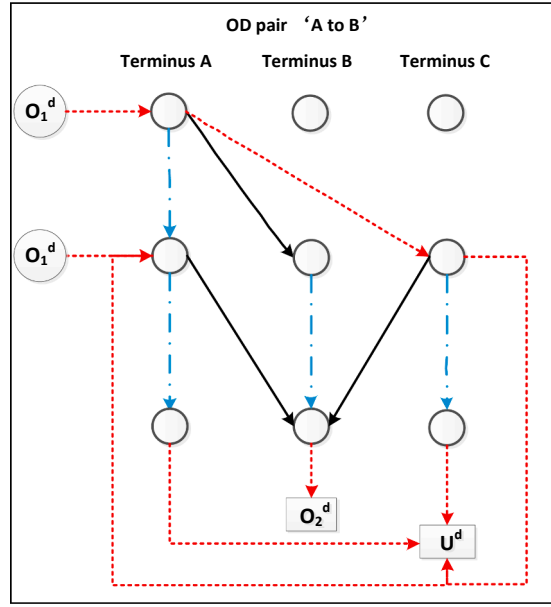


Fig. 3. Illustration of the passenger time-space network.

number of passengers that arrive at the departure terminus whereas the flows on served or lost demand arcs describe the number of served or unserved passengers at the end of the daily service. If serving all demand is important to the bus company, one may set a large penalty for the lost demand arc, so that more services will be provided to carry all the demand, at the expense of a higher operating cost. Let r_d^{jk} be an indicator specifying whether arc (j, k) is connected in the network $G(V^d, A^d)$. The TS networks of passenger flow are constructed as follows:

$$r_d^{jk} = 1 \quad \forall d \in R, j \in O_1^d, k \in T_1^d, (j, k) \in D_2^d \quad (5)$$

$$r_d^{jk} = 1 \quad \forall d \in R, j \in T_2^d, k \in O_2^d, (j, k) \in D_2^d \quad (6)$$

$$r_d^{jk} = 1 \quad \forall d \in R, j \in T^d \setminus T_2^d, k \in U^d, (j, k) \in D_2^d \quad (7)$$

$$r_d^{jk} = 1 \quad \forall d \in R, j \in T^d, (j, k) \in A^d \setminus D_2^d \quad (8)$$

Fig. 1 (right) illustrates the TS network of passenger flow with one OD pair. This network enables passengers to finish their trips in alternative paths. as shown in the small network in Fig. 3. Passengers who want to travel from A to B can either take the direct service from A to B, or detour to C for taking the service from C to B. Note that only the flow to the destination can flow into O_2^d . Otherwise, it will be marked as the lost demand and flow into U^d . Notice that there is no demand for the last few time intervals. Since the end time of the planning horizon is the completion time of the last service trip, the service window for each OD pair will be shorter.

3.1.2. Mathematical formulation

The MD-MVT-VSP based on TSE network can be stated as follows:

(P1)

$$\min_{W, X, Y} \sum_{i \in I} \sum_{o \in O} V_i^1 Y_{io}^{ok} + \sum_{i \in I} \sum_{o \in O} \sum_{(j, k) \in A_{io}} (C_i + E_i) q_i^{jk} Y_{io}^{jk} + \sum_{i \in I, g \in F_i} V_i^2 W_{ig} + \sum_{(j, k) \in A^d \setminus D_2^d} V_s^{jk} \sum_{d \in R} X_d^{jk} + \sum_{(j, k) \in D_2^d, k \in U^d} V_u^{jk} \sum_{d \in R} X_d^{jk} \quad (9)$$

$$\sum_{j: (j, k) \in A_{io}} Y_{io}^{jk} - \sum_{p: (k, p) \in A_{io}} Y_{io}^{kp} = 0 \quad \forall i \in I, o \in O \quad (10)$$

$$\sum_{o \in O} Y_{io}^{ok} \leq K_i \quad \forall i \in I, k \in T \quad (11)$$

$$\sum_{k: (j, k) \in D_2^d} X_d^{jk} = B_d^j \quad \forall j \in O_1^d, d \in R \quad (12)$$

$$\sum_{p_1: (p_1, p_2) \in D_2^d} X_d^{p_1 p_2} + \sum_{k_1: (k_1, k_2) \in D_2^d} X_d^{k_1 k_2} = \sum_{j \in O_1^d} B_d^j \quad \forall p_2 \in O_2^d, k_2 \in U^d, d \in R \quad (13)$$

$$\sum_{j:(j,k) \in A^d} X_d^{jk} - \sum_{p:(k,p) \in A^d} X_d^{kp} = 0 \quad \forall k \in T^d, d \in R \quad (14)$$

$$(1 - \beta)X_d^{jp} = \beta X_d^{jk} \quad \forall j \in T^d \setminus T_2^d, (j, k) \in W^d, (j, p) \in D_2^d, d \in R \quad (15)$$

$$\sum_{d \in R} X_d^{jk} \leq \sum_{i \in I} \sum_{o \in O} Y_{io}^{jk} \zeta_i \quad \forall (j, k) \in S^d \quad (16)$$

$$\sum_{o \in O} \sum_{(j,k) \in A_{io}^\pi \setminus A_{io}^{\pi'}} Y_{io}^{jk} \leq P_{ig} \quad \forall i \in I, g \in F_i, \pi \in \{\pi \mid \pi \in \Pi, \pi - \pi' = \lceil t_i^r / \xi_i \rceil\} \quad (17)$$

$$\frac{1}{\varpi} \sum_{o \in O} \sum_{j:(j,k) \in D_{io}} \sum_{k \in F_{ig}} Y_{io}^{jk} \leq W_{ig} \quad \forall i \in I, g \in F_i \quad (18)$$

$$W \in \text{binary}, X, Y \in \text{integer} \quad (19)$$

Three types of decision variables are set in this problem. Let $Y = \{Y_{io}^{jk}\}$ be the set of integer variables indicating the bus flow from node j to node k in the bus flow network $G(V, A_{io})$. Let $W = \{W_{ig}\}$ be the set of binary variables denoting whether the refueling station g of bus type i is in use. Let $X = \{X_d^{jk}\}$ be the set of integer variables representing the passenger flow from node j to node k for OD pair d in the passenger flow network $G(V^d, A^d)$.

The objective of minimizing the total system cost within the planning horizon is defined in (9), including the operator cost, passenger cost, and external cost of emissions. Let V_i^1 and V_i^2 be the fixed costs of owning each bus and each refueling station for the planning horizon. Let C_i be the operating cost of unit energy consumption associated with the fuel and maintenance costs, and E_i^{jk} be the external cost of emissions correspondingly. For each passenger, let V_s^{jk} be the monetary time cost corresponding to different arcs, where $(j, k) \in A^d \setminus D_2^d$, and V_u^{jk} be the penalty cost on the lost demand arc. The first three terms compose the operator cost and the external cost for the planning horizon, which are: (1) total fixed cost associated with owning all the buses; (2) total trip operating cost and external cost of emissions; and (3) total fixed cost of owning refueling stations. The last two terms refer to the passenger costs, which are: (1) total passenger traveling, waiting, and walking time costs; and (2) lost demand penalty.

As for the constraints, (10) represents the conservations of bus flows at each node k in the bus network $G(V, A_{io})$. Notice that each bus starts and ends its daily service at the same depot is guaranteed when $k \in O$. Constraint (11) ensures that the buses of type i in use should not exceed the maximum allowable fleet size K_i . Constraint (12) assigns B_d^j , passenger demand of node j on OD pair d , to each departure terminus at which its conservations of passenger flows are guaranteed by (13) and (14). (15) indicates that passengers will look for alternative means of transport, such as taxi, after a certain waiting time, where β denotes the lost demand rate for 1 time interval. Let ζ_i be the capacity of bus type i . Constraint (16) requires that the total passenger volume served on arc (j, k) is subject to the total capacity of all types of buses on service arc (j, k) . Let F_{ig} and P_{ig} be the set of time-expanded nodes and the capacity of refueling station g for buses of type i respectively. To ensure the number of refueling buses will not exceed the capacity of refueling station, we make an alternative definition for the arc set A_{io} with respect to time. Let $\Pi = \{1, 2, \dots, \pi\}$ be the set of time intervals in the planning horizon. A_{io}^π represents the set of arcs ended before time π , $\pi \in \Pi$ in $G(V, A_{io})$. Constraint (17) requires that at every time point, the capacity of each refueling station must be satisfied. Constraint (18) defines the use of a refueling station if it is used at least once in all bus networks, where ϖ denotes an extremely large positive number. All in all, the formulation constitutes an ILP, which can solve not only the single fleet scheduling problem, but also the mixed fleet scheduling problem. The proofs are detailed in the following two lemmas.

Lemma 1. The MD-MVT-VSP-TSE formulation is applicable for the single fleet scheduling problem for buses not needing refueling in the planning horizon.

Proof. Suppose $G(V, A_{io})$ is a TSE network of bus type i , where buses have large sufficiently energy capacities without needing to be refueled during the planning horizon, such as diesel buses (DBs). The range and refueling constraints do not need to be incorporated into the VSP, hence energy consumption does not need to be captured in the network. Let us assume that there is only one energy level in each time interval, and this energy level can be any value since energy is not a binding constraint for network generation. Then we generate the network by using the constraints (1)–(4), where the network can be easily simplified into a TS network as follows,

$$u_{io}^{ok} = 1 \quad \forall k \in T \quad (20)$$

$$u_{io}^{jo} = 1 \quad \forall j \in T \cup O' \quad (21)$$

$$u_{io}^{jk} = 1 \quad \forall j \in T \cup F_i \cup O', k \in \{k \mid k \in T \cup O', \tau_i^k = \tau_i^j + t^{jk}\} \quad (22)$$

$$u_{io}^{jf} = 1 \quad \forall j \in \left\{ j \mid j \in T, \tau_i^j \geq \frac{\sigma' Q_i}{\eta_i v} \right\}, f \in \{f \mid f \in F_i, \tau_i^f = \tau_i^j + t^{jf} + t_i^r\} \quad (23)$$

Constraint (23) denotes the refueling visits corresponding to (4), where the connections start from $\sigma' Q_i$. Since there is no energy label in this TS network, we define the starting time for refueling visits by calculating the time for consuming the energy of $\sigma' Q_i$, i.e.

$\frac{\sigma' Q_i}{\eta_i v}$. Note that the TS network is a special case of the TSE network with the refueling time t_i^r to be a parameter unrelated to energy. In fact, we can define the start refueling ratio σ' to be a large value smaller than the safe driving ratio σ . The refueling visits will not occur due to the large energy capacity Q_i . Then we can remove (23) and reformulate (22) as (24),

$$u_{io}^{jk} = 1 \quad \forall j \in T \cup O', k \in \{k | k \in T \cup O', \tau_i^k = \tau_i^j + t^{jk}\} \quad (24)$$

Based on this TS network, which is equivalent to the TSE network with no energy label, we solve **P1**. Since there are no refueling visits, the LHS of constraint (18) is zero, where the decision variables of refueling stations $\mathbf{W} = \{W_{ig}\}$ will also be zero at optimality. Hence, the single fleet scheduling problem without the range and refueling constraints can be solved by the approach of MD-MVT-VSP-TSE in a TS network. $\square$

Lemma 2. The MD-MVT-VSP-TSE formulation is applicable for the mixed fleet scheduling problem involving both buses with refueling need and those without.

Proof. Suppose the set of bus types with different energy sources is $I = \{1\}$. The problem turns into a single fleet scheduling problem. By adopting the approach of MD-MVT-VSP-TSE, either the buses with refueling need or those without can be solved individually.

If there are two or more bus types with different energy sources where $I = \{1, 2, \dots, i\}$, the problem turns into a mixed fleet scheduling problem. Each bus type of the same power source will have its own bus flow network $G(V, A_{io})$. For example, electric buses (EBs) have their own TSE network under range and refueling constraints, whereas the DBs have their TS network without the energy label according to Lemma 1. Let $\mathbf{Y} = \{Y_{EB}^{jk}, Y_{DB}^{jk}\}$ be the set of integer variables indicating the bus flow from node j to node k in the bus flow networks $G(V, A_{EB}^{jk})$ and $G(V, A_{DB}^{jk})$. And there is no candidate refueling station for DBs such that the decision variables of refueling stations are only composed of one bus type, i.e. $\mathbf{W} = \{W_{EB}^{g}\}$. Hence for the EBs network, we can sum up all bus flows starting from one location at one time to the other location at the other time, regardless of the energy levels, which is identical to the TS network for DBs. The same procedure can be applied to constraint (16). Then we can solve **P1**. To sum up, the mixed fleet scheduling problem involving buses with refueling need in the TSE network and buses without refueling need in the TS network can be solved by the approach of MD-MVT-VSP-TSE. $\square$

3.2. MD-MVT-VSP based on the TS network (MD-MVT-VSP-TS)

The TSE network involves three kinds of state variables or labels, which will result in a large augmented network, causing computational issues for larger-scale implementations. Hence, it is advantageous to develop an alternative simplified formulation based on TS network to find approximate solutions for larger-scale problems. Different from the approach in Section 3.1, energy consumption is not tracked explicitly in the alternative formulation, which will greatly reduce the size of the resultant augmented network and hence the required computational effort, as we will see in the numerical studies. Nevertheless, apart from the assumptions in Section 2, one more assumption is added for this simplified formulation: (a) the refueling time of each bus type is fixed, and (b) each bus, with refueling at most once during operation, can fully cover the range of operation of the planning horizon, which is reasonable given the current technology. For example, if each refueling or recharging provides a range of 200 km, with an initially fully fueled or charged bus at the beginning of the planning horizon and one additional refueling or recharging during operation, the bus can have a maximum range of 400 km. By adding this assumption, we can guarantee the feasibility of travel for each bus. In the following, we first introduce the definitions and notation to describe the TS network.

3.2.1. Generation of bus TS network

In the TS network of bus flow, each node represents a specific location at a specific time without tracking the energy consumption. Each network is built by connecting the arcs along a feasible sequence of trips. According to Lemma 1, the TS network is a special case of the TSE network with no energy label, and the refueling time is a parameter unrelated to energy consumption. In this TS network, we define the refueling time t_i^r to be a fixed value for buses of type i . This TS network is generated to cater for buses with refueling need through a set of constraints, as described below. The parameter definitions in this section follow those in Section 3.1.1.1 expect for those associated with depot. In the TS network, all depots are involved in one network without separating individually as each bus has its own network. $A_{io}, S_{io}, W_{io}, D_{io}$ are now defined to be A_i, S_i, W_i, D_i correspondingly. Let u_i^{jk} be an indicator specifying indicate whether arc (j, k) is connected in the network $\hat{G}(V, A_i)$. Each TS network of bus flow $\hat{G}(V, A_i)$ is formulated as: (25)–(28).

$$u_i^{jk} = 1 \quad \forall j \in O, k \in T \quad (25)$$

$$u_i^{jk} = 1 \quad \forall j \in T \cup O', k \in O \quad (26)$$

$$u_i^{jk} = 1 \quad \forall j \in T \cup F_i \cup O', k \in \{k | k \in T \cup O', \tau_i^k = \tau_i^j + t^{jk}\} \quad (27)$$

$$u_i^{jf} = 1 \quad \forall j \in \left\{ j \mid j \in T, \tau_i^j \geq \frac{\sigma' Q_i}{\eta_i v} \right\}, f \in \{f | f \in F_i, \tau_i^f = \tau_i^j + t^{jf} + t_i^r\} \quad (28)$$

3.2.2. Mathematical formulation

By using the same passenger network in Section 3.1.1.2, the MD-MVT-VSP-TS formulation can be stated as follows:

(P2)

$$\min_{\mathbf{Y}, \mathbf{W}, \mathbf{X}} \sum_{i \in I, n \in N_i} \sum_{j \in O} V_i^1 Y_{in}^{jk} + \sum_{i \in I, n \in N_i} \sum_{j, k \in V} (C_i + E_i) q_i^{jk} Y_{in}^{jk} + \sum_{i \in I, g \in F_i} V_i^2 W_{ig} + \sum_{(j, k) \in A_i^d \setminus D_2^d} V_s^{jk} \sum_{d \in R} X_d^{jk} + \sum_{(j, k) \in D_2^d, k \in U^d} V_u^{jk} \sum_{d \in R} X_d^{jk} \quad (29)$$

Passenger network constraints: (5) to (8)

(30)

Bus connectivity constraints:

$$\sum_{j: (j, k) \in A_i} Y_{in}^{jk} - \sum_{p: (k, p) \in A_i} Y_{in}^{kp} = 0 \quad \forall i \in I, n \in N_i, k \in V \quad (31)$$

$$\sum_{n \in N_i} \sum_{o \in O} Y_{in}^{ok} \leq K_i \quad \forall i \in I, k \in T \quad (32)$$

Passenger flow conservation constraints: (12) and (15)

(33)

Capacity constraint:

$$\sum_{d \in R} X_d^{jk} \leq \sum_{i \in I, n \in N_i} Y_{in}^{jk} \zeta_i \quad \forall (j, k) \in S^d \quad (34)$$

Refueling station constraint:

$$\sum_{n \in N_i} \sum_{\substack{k \in F_{ig}, \\ (j, k) \in A_i^{\pi} A_i^{\pi'}}} Y_{in}^{jk} \leq P_{ig} \quad \forall i \in I, g \in F_i, \pi \in \{\pi \mid \pi \in \Pi, \pi - \pi' = \lceil t_i^r / \zeta_i \rceil\} \quad (35)$$

$$\frac{1}{\varpi} \sum_{n \in N_i} \sum_{j: (j, k) \in D_1} \sum_{k \in F_{ig}} Y_{in}^{jk} \leq W_{ig} \quad \forall i \in I, g \in F_i \quad (36)$$

Variable types:

$$\mathbf{W}, \mathbf{Y} \in \text{binary}, \mathbf{X} \in \text{integer} \quad (37)$$

Refueling constraints:

$$\sum_{(j, k) \in A_i^{\pi}} q_i^{jk} Y_{in}^{jk} \leq \sigma Q_i + \sigma Q_i \left(\sum_{f \in F, (j, f) \in A_i^{\pi}} Y_{in}^{jf} \right) \quad \forall i \in I, n \in N_i, \pi \in \Pi \quad (38)$$

$$\pi \leq \frac{(\sigma' + \sigma) Q_i}{\eta_i v \zeta_i} \quad \forall i \in I, \pi \in \Pi \quad (39)$$

Let $N_i = \{1, 2, \dots, K_i\}$ be the index for each individual bus of type i in the network $\hat{G}(V, A_i)$. The decision variables associated with bus flow in this approach change to $\hat{\mathbf{Y}} = \{Y_{in}^{jk}\}$, i.e. the set of binary variables indicating whether bus n of type i travels through arc (j, k) in the bus flow network $\hat{G}(V, A_i)$. The other two variables remain the same in this approach, i.e. $\mathbf{W} = \{W_{ig}\}$ and $\mathbf{X} = \{X_d^{jk}\}$.

Different from the TSE network, energy consumption is not tracked in the TS network. Therefore, we define constraint (38) to capture the refueling need to avoid energy shortage. To begin with, we define A_i^{π} to be the set of arcs ended before time π , $\pi \in \Pi$ in $\hat{G}(V, A_i)$ corresponding to A_{i0}^{π} in (17). Constraint (38) requires that at every time point, the range constraint must be satisfied. For anytime π , $\pi \in \Pi$, the LHS of (38) calculates the cumulative energy consumption of each bus in the network $\hat{G}(V, A_i)$, accounting for all its movements in service arcs, waiting arcs and deadheading arcs at and before time π . The RHS of (38) defines the total energy that a bus in $\hat{G}(V, A_i)$ can use from the beginning to time π by considering the safe driving ratio. The first term on the RHS of (38) denotes the initial energy a bus of type i can use when it starts its daily operations, which we assume to be full with a safe driving ratio σ . The second term in the bracket calculates the total charging times, and we assume that the amount of σQ_i is obtained from each refueling. Hence constraint (38) requires that the cumulative energy consumption should be less than or equal to the sum of the initial energy capacity and the cumulative energy obtained from refueling. It thus guarantees that a refueling action will take place once the total energy consumption is over the safety limit. In this study, the amount of energy gained from each refueling is set to be σQ_i .

Since energy consumption is not tracked explicitly, in some extreme situations, it can happen that certain buses may run out of fuel even if the refueling constraint (38) is satisfied. Theoretically, with an initially fully fueled bus and one additional refueling at a refueling station during operation, the maximum range ought to be related to the fuel content $2\sigma Q_i$. But if refueling occurs too early before complete fuel depletion, the actual range will be shorter. Nevertheless, what can be ensured is that after each refueling, each bus has a range related to σQ_i . Let ξ_t be the duration of each time interval. Constraint (39) ensures that a bus travels within the range no matter how much energy it gains from refueling at a refueling station during operation. We assume that each bus is fully fueled at the start of the planning horizon, and it is refueled after time π_1 , i.e. $\pi_1 = \frac{\sigma Q_i}{\eta_i v}$. Since the bus regains the full fuel capacity after

refueling, it can operate for a duration of π_2 , where $\pi_2 = \frac{\sigma Q_i}{\eta_i v}$. Combining, with an initially fully fueled bus and one refueling during operation, it can be guaranteed that the bus will not run out of fuel for the planning horizon $\pi \leq \frac{(\sigma' + \sigma) Q_i}{\eta_i v \xi_i}$, where π here is expressed in integer multiples of ξ_i . In summary, this alternative formulation is capable of addressing the range and refueling issues, as well as the mixed-route travel problem. Note that in the solution procedure, the time interval can be set to be sufficiently small, and the refueling time can be set to be longer than 1 time intervals. However, if we define the refueling time to be less than or equal to 1 time interval, we obtain the same global optimal solution as that of the MD-MVT-VSP-TSE, as stated in the following proposition. For example, for vehicles with fast recharging, such as battery swapping, then the conditions described in Proposition 1 will most likely be satisfied.

Proposition 1. The two approaches, MD-MVT-VSP-TS and MD-MVT-VSP-TSE, are equivalent when (1) the planning horizon is within twice the travel time range of a fully refueled bus; and (2) the time interval of the network is larger than or equal to the time needed to fully refuel a bus.

Proof. We first discuss the network generation of these two approaches. As we proved in Lemma 1 that the TS network is a special case of the TSE network without considering the energy label, where the time for refueling is a parameter unrelated to energy.

Suppose the time interval of the network is larger than or equal to the time needed for full refueling, then regardless of the form of the refueling time function, or the amount of fuel to be replenished to a bus, the time needed for each refueling will always be 1 time interval. Under this time interval setting, as far as refueling time is concerned, the TS network is equivalent to the TSE network without energy label.

Then we discuss the modal formulation. Let $\widehat{Y}_{in}^{jk}$ be the binary variable indicating that bus n of type i travels through arc (j, k) , where $\sum \widehat{Y}_{in}^{jk} = \sum Y_{io}^{jk}$. Eqs. (1)–(19) can be reformulated by using $\sum \widehat{Y}_{in}^{jk}$ instead of $\sum Y_{io}^{jk}$. Moreover, networks for each depot are developed to guarantee the assumption that buses start and end at the same depot in MVT-VSP-TSE. Nevertheless, all depots are incorporated into one network in MVT-VSP-TS without violating the assumption as each bus has its own network. In summary, under the above time interval setting, we can conclude that (25)–(37) are the same as (1)–(19), by comparing the modal formulation of MVT-VSP-TS with the reformulation of MVT-VSP-TSE.

Finally, we discuss the range and refueling issues. Different from MVT-VSP-TSE which sets up the range and refueling constraints beforehand in the network, as shown in (3) and (4), MD-MVT-VSP-TS solves these issues with the refueling constraints of (38) and (39).

The first constraint (38) is equivalent to (3) and (4) under (39), which indicates two energy rules, i.e. $e_i^j + q_i^{jk} \leq \sigma Q_i, \forall j \in T \cup F_i \cup O'$ and $e_i^j = 0, \forall j \in F_i$. Further proof is given below,

Let e_i^k be the cumulative energy consumption of a type i bus at node k . And all buses are fully refueled at the start of the planning horizon.

Case 1: For any node $k \in \{k | e_i^k \leq \sigma Q_i\}$ before refueling during operation, the cumulative energy consumption of each bus calculated by (3) in MD-MVT-VSP-TSE is:

$$\widehat{e}_i^k = e_i^k = q_i^{jk} \widehat{Y}_{in}^{jk} + e_i^j = q_i^{jk} \widehat{Y}_{in}^{jk} + \left(q_i^{pj} \widehat{Y}_{in}^{pj} + e_i^p \right) = q_i^{jk} \widehat{Y}_{in}^{jk} + \left(q_i^{pj} \widehat{Y}_{in}^{pj} + \dots + 0 \right) = \sum_{(j,k) \in A_i^{jk}} q_i^{jk} \widehat{Y}_{in}^{jk} \leq \sigma Q_i$$

Similarly, the cumulative energy consumption before time $\pi = \tau_i^k$ calculated by (38) in MD-MVT-VSP-TS is:

$$\sum_{(j,k) \in A_i^{jk}} q_i^{jk} Y_{in}^{jk} \leq \sigma Q_i \quad \forall i \in I, n \in N_i$$

Thus, the range constraints for both formulations are the same.

Case 2: For any node $k \in \{k | e_i^k \leq \sigma Q_i\}$ between the first refueling and second refueling during operation, let $k' \in \{k' | e_i^{k'} \leq \sigma Q_i, k' + 1 \in F_i\}$ be the last node before the first refueling visit, the cumulative energy consumption of each bus calculated by (3) and (4) in MD-MVT-VSP-TSE is:

$$\begin{aligned} \widehat{e}_i^k &= e_i^k + e_i^{k'} = \left(q_i^{jk} \widehat{Y}_{in}^{jk} + e_i^j \right) + \left(q_i^{j'k'} \widehat{Y}_{in}^{j'k'} + e_i^{j'} \right) \\ &= \left(q_i^{jk} \widehat{Y}_{in}^{jk} + \left(q_i^{pj} \widehat{Y}_{in}^{pj} + e_i^p \right) \right) + \left(q_i^{j'k'} \widehat{Y}_{in}^{j'k'} + \left(q_i^{p'j'} \widehat{Y}_{in}^{p'j'} + e_i^{p'} \right) \right) \\ &= \left(q_i^{jk} \widehat{Y}_{in}^{jk} + \left(q_i^{pj} \widehat{Y}_{in}^{pj} + \dots + 0 \right) \right) + \left(q_i^{j'k'} \widehat{Y}_{in}^{j'k'} + \left(q_i^{p'j'} \widehat{Y}_{in}^{p'j'} + \dots + 0 \right) \right) \\ &= \left(q_i^{jk} \widehat{Y}_{in}^{jk} + q_i^{pj} \widehat{Y}_{in}^{pj} + \dots + q_i^{qk'} \widehat{Y}_{in}^{qk'} + q_i^{j'k'} \widehat{Y}_{in}^{j'k'} + q_i^{p'j'} \widehat{Y}_{in}^{p'j'} + \dots + 0 \right) \\ &= \sum_{(j,k) \in A_i^{jk}} q_i^{jk} \widehat{Y}_{in}^{jk} \leq \sigma Q_i + e_i^{k'} \end{aligned} \quad (40)$$

Similarly, the cumulative energy consumption of each bus before time $\pi = \tau_i^k$ calculated by (38) in MD-MVT-VSP-TS is:

$$\sum_{(j,k) \in A_i^{jk}} q_i^{jk} Y_{in}^{jk} \leq \sigma Q_i + \sigma Q_i \quad \forall i \in I, n \in N_i, \quad \sum_{k \in F, (j,k) \in A_i^{jk}} Y_{in}^{jk} = 1 \quad (41)$$

Comparing (40) and (41), only the second terms on the RHS are different. According to (39), the time interval in the planning horizon should not exceed $\pi \in \left\{ \pi \mid \pi \leq \frac{(\sigma' + \sigma) Q_i}{\eta_i v_i^k} \right\}$, therefore, the cumulative energy consumption of each bus should be less than or equal to $(\sigma' + \sigma) Q_i$. For (41) of MD-MVT-VSP-TS, we have:

$$\sum_{(j,k) \in A_i^{jk}} q_i^{jk} Y_{in}^{jk} \leq (\sigma' + \sigma) Q_i \quad \forall i \in I, n \in N_i, \quad \sum_{k \in F, (j,k) \in A_i^{jk}} Y_{in}^{jk} = 1$$

According to (28) where $\tau_i^{k'} \geq \frac{\sigma' Q_i}{\eta_i v_i^{k'}}$, we have:

$$\sum_{(j,k) \in A_i^{jk}} q_i^{jk} Y_{in}^{jk} \leq (\sigma' + \sigma) Q_i = \sigma' Q_i + \sigma Q_i \leq \sigma Q_i + \eta_i v_i^{k'} = \sigma Q_i + e_i^{k'}$$

Thus, the range and refueling constraints are the same for both formulations in case 2.

Case 3: According to (39), buses are enabled to travel with the cumulative energy consumption subject to $(\sigma' + \sigma) Q_i$. Thus, a third refueling will not occur at optimality.

In the summary, when the time interval is larger than or equal to the time needed for full refueling, a global optimum can be obtained by the MD-MVT-VSP-TS formulation, which is the same as that of the MD-MVT-VSP-TSE formulation. $\square$

4. Numerical studies

4.1. Small illustrative example

To demonstrate the feasibility and optimality of the solutions, we conduct a small example with four depots, four termini, and four charging stations. Note that **P1** and **P2** are ILPs, and we use IBM ILOG CPLEX Optimization Studio 12.4 without one thread for solutions. The planning horizon consists of six 30-min intervals. For simplicity, we assume that passengers arrive at the beginning of each time interval. As shown in Table 1, there are 3 OD demands. The travel distances of the four OD pairs are shown in Table 2 and the average vehicle speed is 23.5 km/h (Transport Department, 2014). The corresponding adjustment parameters for vehicle mass and speed are not considered in this small example. Four candidate recharging stations for electric vehicles and four candidate depots are set at each terminus.

Li et al. (2012) used the pollution equivalent conversion values in China to estimate the emissions cost, whereas for the Hong Kong case in this paper, we use the external cost following the UK standard. Table 3 shows the bus emissions factors corresponding to the different energy sources, and the external costs of vehicle-related emissions. Notice that we include the indirect emissions from electricity generation for EBs in this study. The attributes of buses and other parameters, such as operating cost, purchasing cost, are shown in Table 4. It is worth noting that the refueling time for the electric bus is assumed to be a linear function with respect to the energy consumption for the first approach, where the function can be relaxed to any form in the future. Define the full refueling time for EBs to be 30 min in this small case, the refueling time function is thus: $f(e_i^j) = \frac{30}{\sigma Q_i} e_i^j = \frac{30}{0.8 \times 230} e_i^j = 0.163 e_i^j$. The fixed cost of owning a bus refers to its purchase cost, whereas that of the refueling station refers to its construction cost. Both of them need to be transformed into the equivalent value of the planning period. For illustration purposes in this small example, we assume that a 40% subsidy is provided to EBs to enhance their financial feasibility. Meanwhile, we assume the energy capacity of EBs to be 16 kWh, and safe driving ratio to be 100% in this small case. For each time interval of each bus terminus, energy consumption is divided into 5 levels by energy step of 4 kWh. According to Li and Ouyang (2011), the amortized annual construction cost of a charging station in China was estimated to be US\$7500. The passenger value of time for 1 time interval (30 min) is US\$ 0.3. The penalty of losing demand is US\$ 10 per passenger, and the lost demand rate β is 0 for simplification.

Two experiments are conducted for this small example by using the two approaches developed in Section 3. The results show that there is a substantial reduction of 70% in the number of variables from MD-MVT-VSP-TSE to MD-MVT-VSP-TS, corresponding to an 83% reduction in computational time. Furthermore, if we take the energy step to be 1 kWh, the reduction of the number of variables is up to 96%. The results obtained from the two approaches are identical as expected. The scheduling results show that one electric bus is needed to carry all the trips with one charging action in the middle. Let the bus termini be represented by A, B, C, D, respectively, depot by O, and refueling station by RS, and the number beside denotes the time interval. For O and RS, their subscripts designate the exact depot or refueling station identity. For example, A1 denotes terminus A at time 1, and RS43 denotes the refueling station #4 at time 3, and O1 depot #1. The path for this electric bus is O₁-A1-B2-D3-RS₄3-RS₄4-D4-C5-O₁, as shown in Fig. 4 (left). By

Table 1
Time-dependent OD demand data of the small example.

OD\time	1	2	3	4	5	6
1(A-B)	72	0	0	0	0	0
2(B-A)	0	0	2	0	0	0
3(C-D)	0	0	0	0	0	0
4(D-C)	0	0	0	70	0	0

Table 2
Travel distance (km) of the small example.

O/D	1	2	3	4
1	0	10	0.8	12
2	10	0	12	0.5
3	0.8	12	0	10
4	12	0.5	10	0

Table 3
Bus emissions factors and the external cost of vehicle-related emissions.

	CO ₂	CO	THC	NO _x
Diesel(g/L) ^a	2600.32	5.57	2.71	24.20
Electric(g/kWh)	867.60 ^b	0.00	0.00	0.00
External cost (US\$/g) ^c	0.000023	0.000520	0.001019	0.003613

^a From Pelkmans et al. (2001).

^b From Doucette and McCulloch (2011).

^c From Cen et al. (2016).

Table 4
Parameters set for different bus types.

	Electric bus	Diesel bus
Lifespan (year)	12 ^a	17 ^b
Energy capacity	230 kWh ^c	350 L ^d
Energy consumption rate	1.2 kWh/km ^c	0.63 L/km ^f
Refueling time(min)	30	10
Passenger capacity	72 ^c	71 ^c
External cost	0.020 US\$/kWh	0.15 US\$/L
Operating cost	0.16 US\$/kWh ^g	1.27 US\$/L ^h
purchase cost (US\$)	790,000 ⁱ	321,143 ^h

^a From Chicago Transit Authority (2017).

^b From Legislative Council of Hong Kong (2014).

^c From The Encyclopedia of Bus Transport in Hong Kong (2016).

^d From Alexander Dennis Limited (2016).

^e From Nylund and Koponen (2012).

^f From Pelkmans et al. (2001).

^g From Noel and McCormack (2014).

^h From Clark et al. (2007).

ⁱ From Dickens et al. (2012).

solving the problem, not only is the charging station located (RS₄ at terminus D), but also is the charging time precisely decided (time 3 to 4). As an essential part in the system cost, the passenger movements are optimized as well. For each OD demand, the passenger flow has three possible directions. Passengers may take the direct service line, wait for the next service trip, or detour to the nearby bus terminus and take a service trip with a similar direction to the destination. Fig. 4 (right) shows all passenger flows for each OD demand. Due to the low cost-effective of launching the bus service from B to A for the demand of 2 passengers, these 2 passengers from B detour from B to D and then C to A using the service trip from D to C.

To further illustrate the solution properties of these two approaches, we conduct a series of experiments with different planning horizons and safe driving ratios. Due to the large problem size associated with the MD-MVT-VSP-TSE formulation, before we apply the formulation to a larger actual Hong Kong scenario, we apply both approaches to a reduced Hong Kong scenario with four termini, i.e. termini E, F, G, and H, whose OD demands, travel distances, and adjustment parameters for vehicle mass and speed are shown in Tables 5–7.

We first vary the planning horizon from 3 to 7 h. The energy levels are divided by an energy step of 2 kWh with a safe driving ratio of 80%, generating 92 nodes in each time interval of each bus terminus. The results are shown in Fig. 3 (left) with double y-axes. The left y-axis is the cost component involving the percentage of the operating, emissions and passenger costs, and the right y-axis is the computational time, where the blue and green curves are associated with the MD-MVT-VSP-TSE and MD-MVT-VSP-TS formulations, respectively. It can be seen from Fig. 5 (left) that all solutions, in terms of optimal costs, are identical by these two approaches as expected, except for the computational time, which is much higher in MD-MVT-VSP-TSE with a longer planning horizon, almost 60 times higher in some cases.

Another key factor that plays an important role in electric bus scheduling is the safe driving ratio. In previous studies of vehicle fleet management, electric vehicles are recharged based on a certain rule of thumb, for example, 80% in Li (2013), i.e., electric

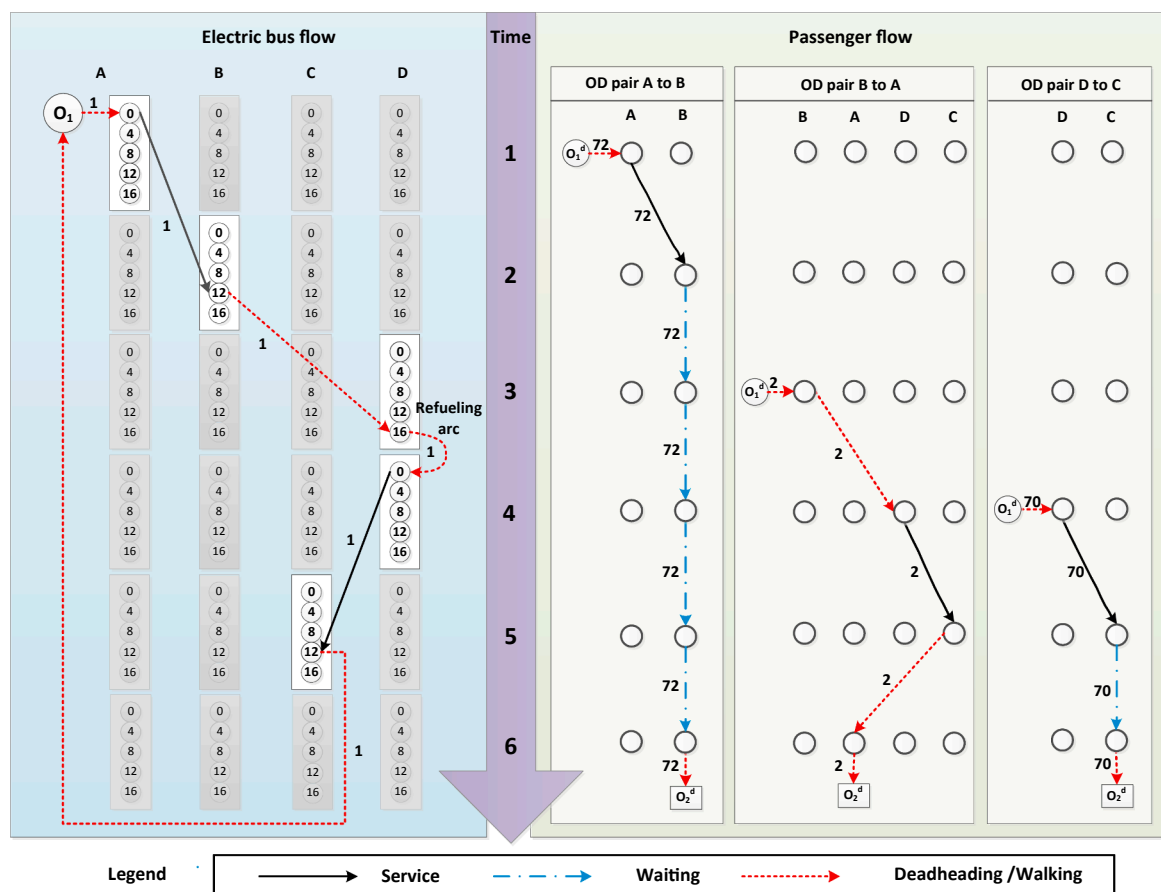


Fig. 4. Optimal bus flow and passenger flow in the small case.

Table 5

Time-dependent OD demand data.

OD/Time	8:00	8:30	9:00	9:30	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30
1(A-B)	20	20	20	20	21	29	56	94	112	97	67	42
2(B-A)	20	20	20	20	21	27	47	76	89	78	55	36
3(C-D)	48	102	179	214	184	124	73	45	34	30	30	30
4(D-C)	10	10	10	10	10	12	19	29	33	29	22	15
5(E-F)	40	40	40	40	40	45	58	77	86	78	63	51
6(F-E)	10	11	17	37	66	79	68	45	26	16	11	10
7(G-H)	20	20	20	20	20	25	38	57	66	58	43	31
8(H-G)	30	30	30	30	30	30	30	31	37	57	86	99
9(I-J)	191	218	195	150	113	91	83	80	80	80	80	80
10(J-I)	10	10	10	10	12	14	15	14	12	11	10	10
11(K-L)	60	60	60	60	61	71	105	153	175	156	118	87
12(L-K)	50	50	50	50	50	55	68	87	96	88	73	61
OD/Time	14:00	14:30	15:00	15:30	16:00	16:30	17:00	17:30	18:00	18:30	19:00	19:30
1(A-B)	28	22	20	20	20	20	20	22	28	0	0	0
2(B-A)	26	21	20	20	20	20	20	21	26	0	0	0
3(C-D)	30	30	34	45	73	124	184	214	179	0	0	0
4(D-C)	12	10	10	10	10	10	10	10	12	0	0	0
5(E-F)	44	41	40	40	40	40	40	41	44	0	0	0
6(F-E)	10	10	10	10	11	16	26	45	68	0	0	0
7(G-H)	24	21	20	20	20	20	20	21	24	0	0	0
8(H-G)	88	65	46	36	31	30	30	30	30	0	0	0
9(I-J)	83	91	113	150	195	218	191	134	94	0	0	0
10(J-I)	10	10	10	10	10	10	10	11	12	0	0	0
11(K-L)	69	62	60	60	60	60	60	62	69	0	0	0
12(L-K)	54	51	50	50	50	50	50	51	54	0	0	0

Table 6
Travel distance (km).

O/D	1	2	3	4	5	6	7	8	9	10	11	12
1	0.0	25.0	4.1	10.5	2.0	13.4	0.8	11.2	3.0	25.6	4.3	15.6
2	26.0	0.0	22.2	11.7	19.8	11.8	18.6	17.6	18.6	2.3	22.2	2.3
3	3.3	22.5	0.0	25.4	1.9	20.6	4.0	11.3	5.5	19.6	0.0	19.6
4	10.6	11.6	25.7	0.0	12.6	5.2	11.4	18.8	14.3	9.6	14.0	9.6
5	2.0	19.8	2.2	13.5	0.0	20.7	1.5	9.8	5.4	17.8	2.5	17.8
6	13.1	9.0	17.1	5.0	21.0	0.0	13.8	19.6	13.7	6.8	17.1	6.8
7	0.9	19.7	3.5	12.1	1.7	14.2	0.0	14.5	2.5	18.0	3.5	18.0
8	11.7	18.0	12.0	16.6	10.3	18.4	15.6	0.0	12.2	18.6	12.0	18.6
9	3.8	20.3	6.2	13.8	4.7	14.4	1.9	11.8	0.0	18.4	6.2	18.4
10	25.6	2.0	19.8	9.5	17.6	8.9	17.1	17.9	19.3	0.0	19.8	0.0
11	3.3	22.5	0.0	14.2	2.7	20.6	4.0	11.3	5.5	19.6	0.0	19.6
12	15.6	2.0	19.8	9.5	17.6	8.9	17.1	17.9	18.4	0.0	19.8	0.0

From the Encyclopedia of Bus Transport in Hong Kong (2016).

Table 7
Adjustment parameters for vehicle mass and speed.

	8:00	8:30	9:00	9:30	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30
γ_m^{jk}	1.0	1.0	1.1	1.1	1.1	1.0	1.0	1.0	1.0	1.0	1.0	1.0
γ_v^{jk}	1.0	1.0	1.1	1.1	1.1	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	14:00	14:30	15:00	15:30	16:00	16:30	17:00	17:30	18:00	18:30	19:00	19:30
γ_m^{jk}	1.0	1.0	1.0	1.0	1.0	1.0	1.1	1.1	1.1	1.0	1.0	1.0
γ_v^{jk}	1.0	1.0	1.0	1.0	1.0	1.0	1.1	1.1	1.1	1.0	1.0	1.0

From Pelkmans et al. (2001).

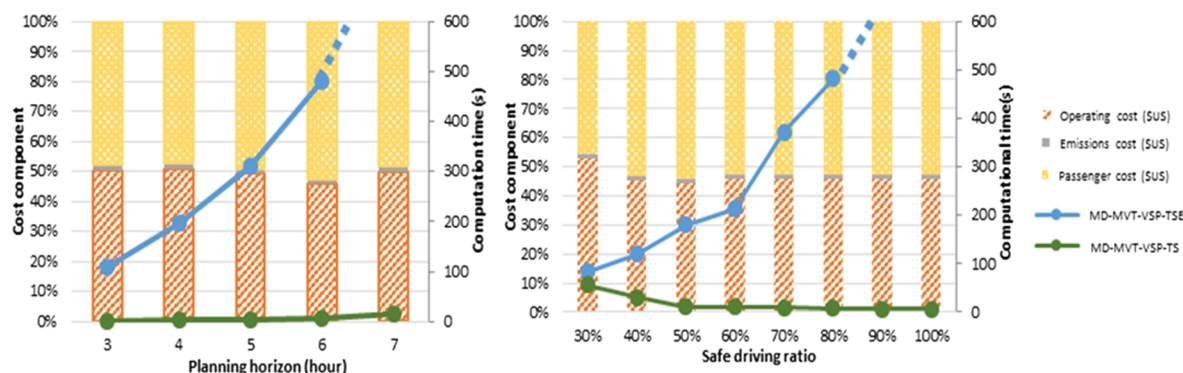


Fig. 5. Cost component and computation time with a larger planning horizon/safe driving ratio.

vehicles are recharged when its travel distance/energy consumption reaches 80% of the maximum travel distance/energy capacity. The question is whether this 80% rule is feasible or efficient. We hence conduct a sensitivity analysis for the safe driving ratio from 30% to 100%, based on the government subsidy ratio of 40%. The results in Fig. 3 (right) show that under the safe driving ratio of 30%, the operator would reduce the passenger cost by purchasing more buses. Yet the bus fleet would return to a regular level when the safe driving ratio is over 40%, with consequential decrease in the cost for passengers.

Note that when the planning horizon is longer than six hours, or when the safe driving ratio is over 80%, the problem size of the MD-MVT-VSP-TSE formulation becomes too big, and the computer ran out of memory and could not finish the solution procedure. We conduct another experiment to investigate the impacts of energy step size on the solution, as shown in Fig. 6. On one hand, the computation time decreases with a larger energy step size. On the other hand, it is interesting to find out that the system cost does not increase monotonically, due to the energy step discretization, but generally shows an upward trend. In this section, we select the smallest energy step size of 2 kWh, as the problem cannot be solved when the energy step is 1 kWh due to the large problem size.

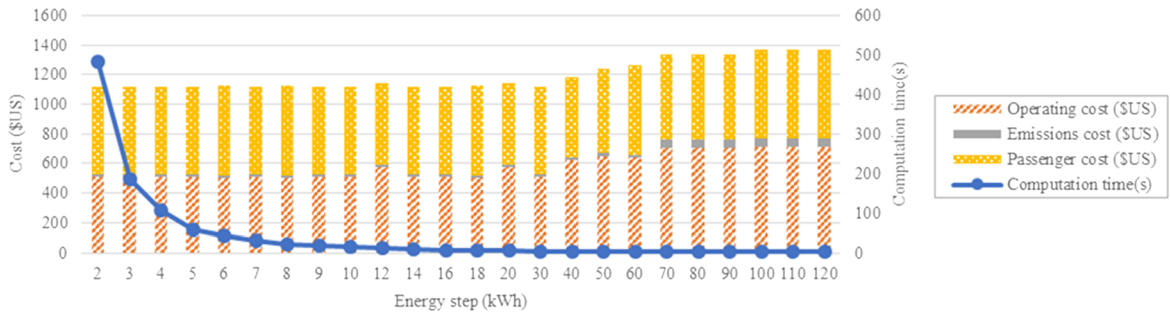


Fig. 6. Cost component and computation time with the change of energy step.

4.2. Application to an actual bus network in Hong Kong

4.2.1. Network data acquisition

Having shown that the simplified MD-MVT-VSP-TS formulation, which does not capture energy consumption explicitly, can produce good solutions as compared with the full-blown MD-MVT-VSP-TSE formulation, we apply the simplified MD-MVT-VSP-TS formulation to an actual Hong Kong bus network, which is too large for the full-blown MD-MVT-VSP-TSE formulation. This network contains all the routes, a total of six, operated by Hong Kong New World First Bus Company in Tseung Kwan O, Hong Kong. We thus consider the problem with 12 bus termini for a 12-h daytime (8:00–20:00), partitioned into 30-min intervals, as shown in Fig. 7. Each OD pair has 24 time-dependent OD demands, where altogether there are 288 time-dependent OD demands, as shown in Table 5. We define the end time of the planning period to be the finish time of the last service trip, which results in a number of zero OD demands near the end time. The travel distances of all the OD pairs are shown in Table 6, where the corresponding adjustment parameters for vehicle mass and speed are shown in Table 7. Two candidate recharging stations for electric vehicle and two candidate depots are allocated at the departure termini of the first and seventh OD pairs (A and G); this assumption can be easily relaxed in future work. The maximum allowable fleet size is set to be 40 for EBs, where the DBs adopt the simplified TS network as stated in Lemma 1. Other parameters follow the same estimation in Section 4.1.

4.2.2. Results

In this section, we analyze the optimal mixed bus fleet scheduling scheme for system cost minimization under the safe driving ratio σ of 80%. We set the maximum running time to be 10 h in this case. The results show that 21 EBs and 14 DBs are needed to carry all demands, with the optimality gap of 3.19%. The optimal schedules are shown in Table 8. The first column provides the bus types. The travel paths for the buses are shown in the second column, where the alphabet represents the bus terminus and the number denotes the time interval. Take the first electric bus for example, it departs from and ends at depot 1. It first serves the OD pair between L and K before carrying the remaining service trips between G and H to the end of the day. At time 16, it goes for charging at charging station 2. By calculating the proportion of service time in the planning horizon, the utilization rate of each bus is calculated in column three, with an average rate of 63%. By comparing with the single-route scheduling, we are pleasantly surprised to find out that the mixed-route scheduling produces far better results, with an 5% improvement of utilization rate. Meanwhile, the fleet size is largely reduced by 24%, saving 8% operating costs and 24% emissions costs. Moreover, the time-varying demand has a noticeable influence on the resultant schedule. Take the bus deployment of services between K and L to illustrate, every 30 min, two or more buses are assigned to serve during peak hours whereas only one bus is needed in the rest of the time. If one were to make the schedule regular throughout the day, say one dispatch per 30 min, there will be either redundant or insufficient supplies, resulting in at least 65% performance loss. In summary, with the increase of the network size, we anticipate this benefit brought by mixed-route scheduling could be even greater.

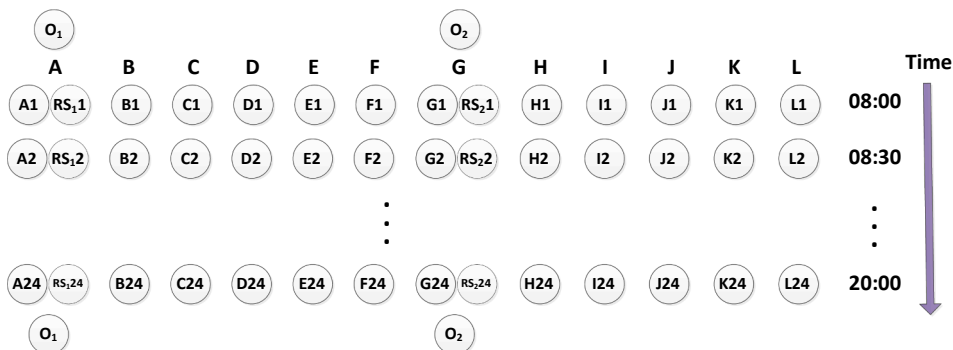


Fig. 7. Nodes in the TS network of bus flow in the Hong Kong case study.

Table 8

Optimal bus scheduling scheme without government subsidy under the safe driving ratio of 80%

Bus type	Path	Utilization rate
Electric	O ₁ -L2-K4-L6-K8-L10-K12-H13-G15-G16-RS ₂ 16-RS ₂ 17-G17-H19-G21-H23-O ₁	75%
	O ₁ -I3-J5-F6-E8-F10-E12-F14-E16-RS ₁ 17-RS ₁ 18-C19-D22-O ₁	63%
	O ₁ -C2-D5-A6-B9-A12-RS ₁ 12-RS ₁ 13-I14-J16-I18-J20-O ₁	63%
	O ₁ -H2-G4-C5-D8-A9-B12-J13-I15-J17-RS ₁ 19-RS ₁ 20-C21-D24-O ₁	63%
	O ₁ -I4-J6-I8-J10-I12-J14-RS ₁ 16-RS ₁ 17-H18-G20-C21-D24-O ₁	63%
	O ₁ -I2-J4-D5-A6-K7-L9-K11-L13-RS ₁ 15-RS ₁ 16-E17-F19-E21-F23-O ₁	58%
	O ₁ -K2-L4-K6-L8-K10-K11-RS ₂ 12-RS ₂ 13-K14-L16-K18-L20-K22-L24-O ₁	75%
	O ₁ -C2-D5-C8-D11-C14-D17-RS ₁ 18-RS ₁ 19-C20-D23-O ₁	75%
	O ₁ -E2-F4-E6-F8-E10-F12-F13-RS ₁ 15-RS ₁ 16-I17-J19-I21-J23-O ₁	67%
	O ₁ -I3-J5-I7-J9-I11-J13-RS ₁ 15-RS ₁ 16-K17-L19-K21-L23-O ₁	67%
	O ₁ -D1-C4-D7-A8-B11-A14-RS ₁ 14-RS ₁ 15-K16-L18-K20-L22-O ₁	75%
	O ₁ -G4-H6-G8-K9-L11-K13-L15-K17-RS ₂ 18-RS ₂ 19-C20-D23-O ₁	63%
	O ₁ -J1-B2-A5-C6-D9-A10-B13-A16-RS ₁ 16-RS ₁ 17-I19-I20-J22-O ₁	58%
	O ₁ -I1-J3-I5-J7-I9-J11-RS ₁ 13-RS ₁ 14-A14-B17-D18-C21-D24-O ₁	79%
	O ₁ -I1-J3-I5-J7-B8-A11-B14-A17-RS ₁ 17-RS ₁ 18-C19-D22-O ₁	75%
	O ₁ -I2-J4-F5-E7-F9-E11-F13-RS ₁ 15-RS ₁ 16-I17-J19-I21-J23-O ₁	67%
	O ₁ -H1-G3-I4-J6-I8-J10-I12-J14-I16-RS ₁ 17-RS ₁ 18-A18-B21-A24-O ₁	83%
	O ₂ -K1-L3-K5-L7-K9-L11-L12-RS ₁ 14-RS ₁ 15-A15-B18-A21-B24-O ₂	79%
	O ₂ -E1-F3-E5-F7-E9-F11-RS ₁ 13-RS ₁ 14-I15-J17-I19-J21-I23-O ₂	75%
	O ₂ -C3-D6-G7-K8-L10-K12-L14-RS ₁ 16-RS ₁ 17-C18-D21-C24-O ₂	63%
	O ₂ -K10-L12-K14-E15-F17-E19-F21-E23-O ₂	50%
Diesel	O ₁ -A3-B6-A9-B12-A15-A16-A17-A18-I19-J21-O ₁	58%
	O ₁ -C4-D7-G8-H10-G12-H14-G16-I17-J19-O ₁	54%
	O ₁ -C5-D8-C11-D14-D15-D16-C19-D22-O ₁	63%
	O ₁ -E3-F5-F6-E8-C9-D12-D13-C16-C17-D20-O ₁	54%
	O ₁ -I2-J4-I6-J8-I10-J12-I14-I15-I16-I17-I18-J20-O ₁	58%
	O ₁ -C3-D6-G7-H9-G11-H13-G15-I16-J18-I20-J22-O ₁	71%
	O ₁ -L1-K3-K4-K5-K6-K7-K8-K9-L11-L12-L13-K15-L17-K19-L21-O ₁	50%
	O ₂ -C4-D7-A8-G9-H11-G13-C14-C15-C16-C17-D20-O ₂	42%
	O ₂ -C5-D8-A9-K10-L12-B13-B14-B15-A18-E19-E20-F22-O ₂	42%
	O ₂ -C6-D9-C12-C13-D16-A17-I18-J20-O ₂	46%
	O ₂ -G2-H4-G6-H8-G10-H12-G14-H16-G18-G19-H21-G23-O ₂	83%
	O ₂ -I1-J3-B4-A7-B10-A13-A14-A15-E16-F18-F19-F20-E22-O ₂	63%
	O ₂ -I3-J5-J6-B7-A10-A11-I12-I13-J15-I17-J19-B20-L21-K23-O ₂	54%
	O ₂ -K3-L5-K7-L9-K11-L13-L14-K16-E17-E18-C19-C20-D23-O ₂	63%

4.3. Discussion on government policy to promote electric bus

4.3.1. Subsidy on purchasing EBs

In this study, we include CO₂ emissions for EBs from power generation. Despite its emissions incurred indirectly from electricity generation, electric bus is considered as a zero-emission transport mode in view of its roadside emissions. Meanwhile, the serious consequence of roadside emissions may be underestimated. It is reported that 11,490 deaths were caused by heavily NO₂ pollution in England in 2013 (European Environmental Protection Agency, 2016). While in Hong Kong in 2016, there were 1508 premature deaths from air pollution, directly impacting economic losses of up to US\$ 2.3 billion (Hedley Environmental Index, 2016). Therefore, it is necessary to promote the use of EBs. In 2016, the Hong Kong government funded US\$23 million to subsidize franchised bus companies to purchase 36 single-deck EBs (Legislative Council of Hong Kong, 2016), where the subsidy ratio was up to 80%. We thus investigate the impact of government subsidy on EBs in this section.

Five different subsidy ratios on purchasing electric buses varying from 10% to 50% are modeled to investigate the impact on bus fleet structure and different cost components, as shown in Fig. 8 (left). The results show that a higher government subsidy leads to a higher replacement of DBs by EBs, as expected. For this case, the amount of emissions, especially NO_x, drops sharply owing to the introduction of EBs, as shown in Fig. 8 (right). Since there is an over 95% reduction in the amounts of NO_x, at least 1433 lives and US \$ 2.2 billion will be saved for better roadside air quality (Hedley Environmental Index, 2016).

4.3.2. Toll charge for Low Emission Zone (LEZ)

LEZ involves an emission charging scheme with the aim of encouraging the use of cleaner vehicles, which has been operated in many cities in the world, such as London, Hong Kong, etc. Vehicles that do not comply with the stipulated emission standards are charged a toll, whereas vehicles that comply with the emission standards enter the LEZ free of charge. According to the Hong Kong government, since franchised buses accounted for up to 40% of traffic volume in the CBD, three LEZs were designed for franchised buses by the end of 2015 to improve roadside air quality (Legislative Council of Hong Kong, 2014). Few studies have explicitly considered the bus scheduling problem under LEZ; such problems can be properly analyzed using the two approaches developed in this study, as discussed below considering the toll charge design for LEZ.

We consider that the toll charge of LEZ is changeable based on time of day. Let ψ_i^{jk} be the toll for bus type $i \in I$ travelling on arc $(j,$

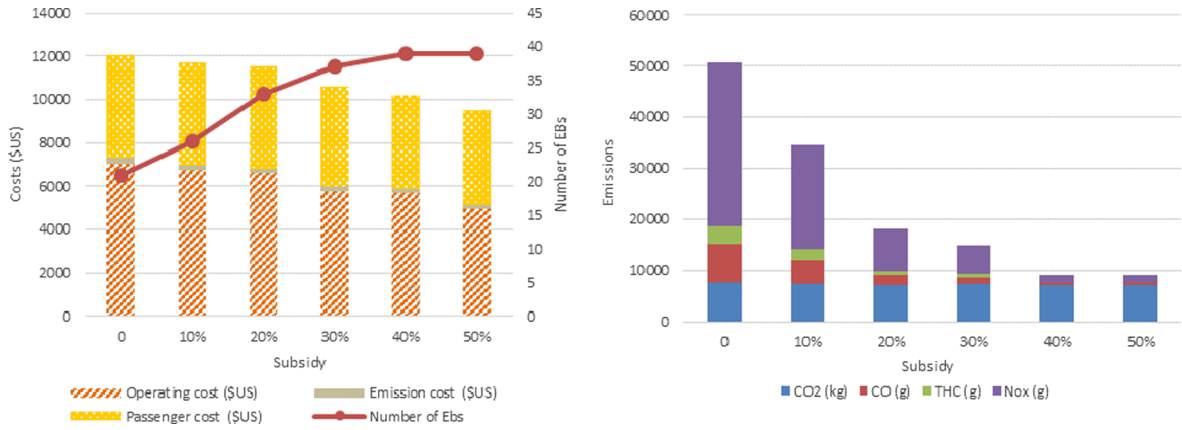


Fig. 8. Costs and emissions with the change of subsidy ratio.

k) in the LEZ. The problems can be reformulated as **P3** and **P4** associated with the approaches of MD-MVT-VSP-TSE and MD-MVT-VSP-TS, respectively.

(P3)

$$\begin{aligned} \min_{w, X, Y} \quad & \sum_{i \in I} \sum_{o \in O} V_i^1 Y_{io}^{ok} + \sum_{i \in I} \sum_{o \in O} \sum_{(j,k) \in A_{io}} ((C_i + E_i) q_i^{jk} + \psi_i^{jk}) Y_{io}^{jk} + \sum_{i \in I, g \in F_i} V_i^2 W_{ig} + \sum_{(j,k) \in A^d \setminus D_2^d} V_s^{jk} \sum_{d \in R} X_d^{jk} \\ & + \sum_{(j,k) \in D_2^d, k \in U^d} V_u^{jk} \sum_{d \in R} X_d^{jk} \end{aligned} \quad (42)$$

s.t. (1) to (8) and (10) to (19)

(P4)

$$\begin{aligned} \min_{Y, W, X} \quad & \sum_{i \in I, n \in N_i} \sum_{j \in O} V_i^1 Y_{in}^{jk} + \sum_{i \in I, n \in N_i} \sum_{j,k \in V} ((C_i + E_i) q_i^{jk} + \psi_i^{jk}) Y_{in}^{jk} + \sum_{i \in I, g \in F_i} V_i^2 W_{ig} + \sum_{(j,k) \in A^d \setminus D_2^d} V_s^{jk} \sum_{d \in R} X_d^{jk} \\ & + \sum_{(j,k) \in D_2^d, k \in U^d} V_u^{jk} \sum_{d \in R} X_d^{jk} \end{aligned} \quad (43)$$

s.t. (25) to (28) and (30) to (39)

Four OD pairs (E-F, F-E, G-H, H-G) are assumed to travel through the CBD, where the periods of 8:00 to 9:00 and 13:30 to 14:30 are defined to be the Low Emission Zone (LEZ). A set of experiments are conducted to investigate the lowest toll charge for turning the LEZ into a Zero Emission Zone (ZEZ). The results show that when the toll charge is over US\$9 per entry, the LEZ will become ZEZ. No DBs will enter the zone due to the sufficiently high toll charge. For this case, we take the travel path of an electric bus to illustrate the bus scheduling under ZEZ, i.e. O₁-C4-D7-G8-H10-**G12-H14-G16-RS₂16-RS₂17-C20-D23-O₁**, where the ones in bold denote the trips travelling through the ZEZ. In summary, other than government subsidy, implementing toll charges for LEZ is another effective measure to improve roadside emissions.

5. Concluding remarks

In this paper, we formulated the multiple depot vehicle scheduling problem with multiple vehicle types (MD-MVT-VSP) under range and refueling constraints. Bus service coordination and the external cost of emissions were taken into consideration to generate a cost-effective and environmentally friendly bus scheduling scheme. To precisely solve the range and refueling issues, we developed a novel approach to generate the feasible time-space-energy (TSE) networks for each vehicle type. Likewise, a time-space (TS) network of passenger flow was generated to represent the passenger movements. Based on these two kinds of feasible flow networks, an integer linear program (ILP) was developed to find the global optimal solution, which gives (1) the bus fleet size and its composition; (2) optimal service deployment schedule depicting the trip schedule as well as the vehicle schedule; (3) locations of the refueling stations; (4) route of each bus including the refueling activities; (5) passenger movements; and (6) total system cost including the operator cost, passenger cost, and the external cost of emissions. For computational efficiency, we developed a simplified formulation based on TS bus flow network, namely MD-MVT-VSP-TS, to find approximate solutions for larger-scale problems, where the computational time was substantially reduced as compared with the former approach. Furthermore, we proved that global optimality was achieved by using the MD-MVT-VSP-TS when (1) the planning horizon is within twice the travel time range of a fully refueled bus; and (2) the time interval of the network is larger than or equal to the time needed to fully refuel a bus.

We successfully applied the MD-MVT-VSP model to an actual bus network in Hong Kong. The results showed that the utilization rate for buses was greatly improved due to mixed-route scheduling. As far as government subsidy for EBs is concerned, the results showed that the amount of emissions was greatly reduced due to the introduction of EBs. This study also demonstrated the

effectiveness of applying the formulations developed to address the bus scheduling problem for Low Emission Zone with toll charges.

Furthermore, we conducted sensitivity analyses on the safe driving ratio of EBs. The energy consumption or the related driving range of EBs is subject to a lot of uncertainty in actual operations, which will have important implications on the scheduling problem, as well as on the risk of bus stranding. We are currently working on how to take into account such stochastic variation on the scheduling problem. In this study, we consider the demand to be deterministic. But the interactions between the supply and demand patterns can be further explored. In future, we will investigate the bus scheduling problem under demand uncertainty. We hope that the formulations developed in this study will form a platform to investigate these important issues.

Acknowledgements

The study is supported by Research Grant SBI13EG04-B, General Research Funds #16206114, and #16211217 of the Research Grants Council of the HKSAR Government, National Natural Science Foundation #71622007, #71861167001 of China, Strategic Research Funding Initiative SRFI11EG15.

Appendix A. Notation

Parameters

Sets

Sets corresponding to the bus flow network

A_{io}/A_i	Set of arcs in the bus flow network $G(V, A_{io})/\hat{G}(V, A_i)$
A_{io}^π/A_i^π	Set of arcs ended before time π , $\pi \in \Pi$ in the bus flow network $G(V, A_{io})/\hat{G}(V, A_i)$
D_{io}/D_i	Set of deadheading arcs in the bus flow network $G(V, A_{io})/\hat{G}(V, A_i)$
F	Set of time-expanded refueling stations
I	Set of bus types with different energy sources, $I = \{1, 2, \dots, i\}$
N	Set of indexes indicating the specific bus of type i travelled in the network $\hat{G}(V, A_i)$, $N = \{1, 2, \dots, n\}$
O	Set of depots
O'	Set of dummy nodes for each depot indicating the temporary stay of buses at depot
S_{io}/S_i	Set of service arcs in the bus flow network $G(V, A_{io})/\hat{G}(V, A_i)$
T	Set of time-expanded bus termini
V	Set of nodes with $V = O \cup T \cup F \cup O'$
W_{io}/W_i	Set of wait arcs in the bus flow network $G(V, A_{io})/\hat{G}(V, A_i)$
Π	Set of time intervals in the planning horizon, $\Pi = \{1, 2, \dots, \pi\}$

Sets corresponding to passenger flow network

A^d	Set of arcs in the dth passenger flow network (for demand on OD pair d)
D^d	Set of detouring arcs in the dth passenger flow network (for demand on OD pair d)
D_1^d	Set of walking arcs in the dth passenger flow network (for demand on OD pair d)
D_2^d	Set of demand arcs in the dth passenger flow network (for demand on OD pair d)
O^d	Set of demands in the dth passenger flow network (for demand on OD pair d)
O_1^d	Set of time-expanded original demands at the beginning of the daily service in the dth passenger flow network (for demand on OD pair d)
O_2^d	Set of served demands at the end of the daily service in the dth passenger flow network (for demand on OD pair d)
R	Set of OD pairs
S^d	Set of service arcs in the dth passenger flow network (for demand on OD pair d)
T^d	Set of time-expanded bus termini in the dth passenger flow network (for demand on OD pair d)
T_1^d	Set of time-expanded departure bus termini of OD pair d, $d \in R$
T_2^d	Set of time-expanded destination bus termini of OD pair d, $d \in R$
U^d	Set of unserved demand at the end of the daily service in the dth passenger flow network (for demand on OD pair d)
V^d	Set of nodes in the dth passenger flow network (for demand on OD pair d)
W^d	Set of wait arcs in the dth passenger flow network (for demand on OD pair d)

Input parameters

B_d^{jk}	Demand of OD pair d on link (j, k)
C_i	Unit bus operating cost of type i
d^{jk}	Travel distance from j to k
e_i^j	Energy consumption at node j in the bus flow network $G(V, A_{io})/\hat{G}(V, A_i)$
$\sim_k$	Cumulative energy consumption of buses of type i travelled to node k
e_i	Unit external cost of emissions of bus type i travelled on link (j, k)
E_i^{jk}	Unit external cost of emissions of bus type i travelled on link (j, k)
$G(V, A_{io})$	Bus time-space-energy network for buses of type i
$\hat{G}(V, A_i)$	Bus time-space network for buses of type i
i	Bus type with different energy sources
(j, k)	Feasible arc linking node j and k
K_i	Maximum allowable fleet size of buses of type i
P_{ig}	Capacity of refueling station g for buses of type i
Q_i	Energy capacity of buses of type i

q_i^{jk}	Energy consumption of bus type i travelled on link(j, k)
r_d^{jk}	Indicator specifying whether arc (j, k) is in the passenger flow network $G(V^d, A^d)$
t^{jk}	Travel time from node j to k
t_i^r	Refueling time of buses of type i
u_{io}^{jk}/u_i^{jk}	Indicators specifying whether arc (j, k) is in the bus network $G(V, A_{io})/\widehat{G}(V, A_i)$
v	Average travel speed
V_i^1	Unit bus owning cost of type i
V_i^2	Unit refueling station owning cost of type i
V_u^{jk}	Penalty of lost demand on link(j, k)
V_s^{jk}	Passenger value of time who travelled on link(j, k)
β	Demand lost rate
γ_m^{jk}	Adjustment parameter for vehicle mass
γ_v^{jk}	Adjustment parameter for speed
ζ_i	Bus capacity of type i
η_i	Energy consumption rate of buses of type i
ξ_e	Energy step
ξ_t	Duration of the time interval
σ	Safe driving ratio
σ'	Start refueling ratio
τ_i^j	Time of arrival at node j in the bus flow network $G(V, A_{io})/\widehat{G}(V, A_i)$
ψ_i^{jk}	Entering price of LEZ for the bus of type i travelling on arc(j, k)
ϖ	Extremely large positive number
Decision variables	
$\mathbf{W}$	Set of binary decision variables indicating if the refueling station g for bus type i is in use, $\mathbf{W} = \{W_{ig}\}$
$\mathbf{X}$	Set of integer decision variables indicating the passenger flow from node j to node k for OD pair d , $\mathbf{X} = \{X_d^{jk}\}$
$\mathbf{Y}$	Set of integer decision variables indicating the bus flow travelled on link (j, k) in the bus flow network $G(V, A_{io})$, $\mathbf{Y} = \{Y_{io}^{jk}\}$
$\widehat{\mathbf{Y}}$	Set of binary decision variables indicating whether bus n of type i travels through arc (j, k) in the bus flow network $\widehat{G}(V, A_i)$, $\widehat{\mathbf{Y}} = \{\widehat{Y}_{in}^{jk}\}$

Appendix B. Acronyms

CNG	compressed natural gas
DBs	diesel buses
EBs	electric buses
ILP	integer linear program
LEZ	low emission zone
LHS	left-hand side
MD	multiple depot
MD-MVT-VSP	multiple depot vehicle scheduling problem with multiple vehicle types
MD-MVT-VSP-TS	multiple depot vehicle scheduling problem with multiple vehicle types based on the time-space network
MD-MVT-VSP-TSE	multiple depot vehicle scheduling problem with multiple vehicle types based on the time-space-energy network
MVT	multiple vehicle types
OD	origin-destination
RHS	right-hand side
TS	time-space
TSE	time-space-energy
VPS	vehicle scheduling problem
ZEZ	zero emission zone

References

- Alexander Dennis Limited (ADL), 2016. <http://www.alexander-dennis.com/products/enviro500-asia-europe/>.
- An, K., Lo, H.K., 2014. Ferry service network design with stochastic demand under user equilibrium flows. *Transp. Res. Part B: Methodol.* 66, 70–89.
- An, K., Lo, H.K., 2015. Robust transit network design with stochastic demand considering development density. *Transp. Res. Part B: Methodol.* 81, 737–754.
- An, K., Lo, H.K., 2016. Two-phase stochastic program for transit network design under demand uncertainty. *Transp. Res. Part B: Methodol.* 84, 157–181.
- Beltran, B., Carrese, S., Cipriani, E., Petrelli, M., 2009. Transit network design with allocation of green vehicles: a genetic algorithm approach. *Transp. Res. Part C: Emerg. Technol.* 17 (5), 475–483.
- Bunte, S., Klierer, N., 2009. An overview on vehicle scheduling models. *Public Transp.* 1 (4), 299–317.
- Ceder, A., Hassold, S., Dunlop, C., Chen, I., 2013. Improving urban public transport service using new timetabling strategies with different vehicle sizes. *Int. J. Urban Sci.* 17 (2), 239–258.
- Cen, X., Lo, H.K., Li, L., 2016. A framework for estimating traffic emissions: the development of Passenger Car Emission Unit. *Transp. Res. Part D: Transp. Environ.* 44, 78–92.
- Chen, Z., He, F., Yin, Y., 2016. Optimal deployment of charging lanes for electric vehicles in transportation networks. *Transp. Res. Part B: Methodol.* 91, 344–365.
- Chicago Transit Authority, 2017. <http://www.transitchicago.com/electricbus/>.

- Clark, N.N., Zhen, F., Wayne, W.S., Lyons, D.W., 2007. Transit bus life cycle cost and year 2007 emissions estimation (No. FTA-WV-26-7004.2007. 1).
- Dickens, M., Neff, J., Grisby, D., 2012. APTA 2012 Public Transportation Fact Book.
- Doucette, R.T., McCulloch, M.D., 2011. Modeling the CO₂ emissions from battery electric vehicles given the power generation mixes of different countries. *Energy Policy* 39 (2), 803–811.
- European Environmental Protection Agency, 2016. <https://www.eea.europa.eu/>.
- Fusco, G., Alessandrini, A., Colombaroni, C., Valentini, M.P., 2013. A model for transit design with choice of electric charging system. *Proc.-Social Behav. Sci.* 87, 234–249.
- Goeke, D., Schneider, M., 2015. Routing a mixed fleet of electric and conventional vehicles. *Eur. J. Operat. Res.* 245 (1), 81–99.
- Hassold, S., Ceder, A., 2012. Multiobjective approach to creating bus timetables with multiple vehicle types. *Transp. Res. Rec.: J. Transp. Res. Board* 2276, 56–62.
- Hassold, S., Ceder, A., 2014. Public transport vehicle scheduling featuring multiple vehicle types. *Transp. Res. Part B: Methodol.* 67, 129–143.
- He, F., Wu, D., Yin, Y., Guan, Y., 2013. Optimal deployment of public charging stations for plug-in hybrid electric vehicles. *Transp. Res. Part B: Methodol.* 47, 87–101.
- Hedley Environmental Index, 2016. hedleyindex.sph.hku.hk. The University of Hong Kong.
- Kim, M.E., Schonfeld, P., 2014. Integration of conventional and flexible bus services with timed transfers. *Transp. Res. Part B: Methodol.* 68, 76–97.
- Kliwer, N., Gintner, V., Suhl, L., 2008. Line change considerations within a time-space network based multi-depot bus scheduling model. In: *Computer-Aided Systems in Public Transport*. Springer, Berlin Heidelberg, pp. 57–70.
- Kliwer, N., Mellouli, T., Suhl, L., 2006. A time-space network based exact optimization model for multi-depot bus scheduling. *Eur. J. Operat. Res.* 175 (3), 1616–1627.
- Legislative Council of Hong Kong, 2014. LCQ14: Establishment of low emission zones for buses. <http://www.info.gov.hk/gia/general/201411/12/P201411120473.htm>. Hong Kong Special Administrative Region.
- Legislative Council of Hong Kong, 2016. LCQ17: Promoting the wider use of electric buses. <http://www.info.gov.hk/gia/general/201601/27/P201601270386.htm>. Hong Kong Special Administrative Region.
- Lai, M.F., Lo, H., 2004. Ferry service network design: optimal fleet size, routing, and scheduling. *Transp. Res. Part A: Policy Practice* 38, 305–328.
- Li, J.Q., Head, K.L., 2009. Sustainability provisions in the bus-scheduling problem. *Transp. Res. Part D: Transp. Environ.* 14 (1), 50–60.
- Li, J.Q., 2013. Transit bus scheduling with limited energy. *Transp. Sci.* 48 (4), 521–539.
- Li, L., Cai, M., Liu, Y., 2012. Integrated benefit evaluation of pedestrian bridge. *Environ. Model. Assess.* 17 (3), 301–313.
- Li, L., Lo, H.K., Cen, X., 2015. Optimal bus fleet management strategy for emissions reduction. *Transp. Res. Part D: Transp. Environ.* 41, 330–347.
- Li, L., Lo, H.K., Xiao, F., Cen, X., 2018. Mixed bus fleet management strategy for minimizing overall and emissions external costs. *Transp. Res. Part D: Transp. Environ.* 60, 104–118.
- Li, Z., Ouyang, M., 2011. The pricing of charging for electric vehicles in China—Dilemma and solution. *Energy* 36 (9), 5765–5778.
- Lim, S., Kuby, M., 2010. Heuristic algorithms for siting alternative-fuel stations using the flow-refueling location model. *Eur. J. Operat. Res.* 204 (1), 51–61.
- Liu, H., Wang, D.Z., 2017. Locating multiple types of charging facilities for battery electric vehicles. *Transp. Res. Part B: Methodol.* 103, 30–55.
- Lo, H.K., An, K., Lin, W.H., 2013. Ferry service network design under demand uncertainty. *Transp. Res. Part E: Logistics Transp. Rev.* 59, 48–70.
- Noel, L., McCormack, R., 2014. A cost benefit analysis of a V2G-capable electric school bus compared to a traditional diesel school bus. *Appl. Energy* 126, 246–255.
- Nylund, N.-O., Koponen, K., 2012. Fuel and technology alternatives for buses. *VTT Technol.* 46.
- Pelkmans, L., De Keuleleere, D., Lenaers, G., 2001. Emissions and fuel consumption of natural gas powered city buses versus diesel buses in real-city traffic. VITO—Flemish Institute for Technological Research (Belgium).
- Pternea, M., Kepaptsoglou, K., Karlaftis, M.G., 2015. Sustainable urban transit network design. *Transp. Res. Part A: Policy Practice* 77, 276–291.
- Schneider, M., Stenger, A., Goeke, D., 2014. The electric vehicle-routing problem with time windows and recharging stations. *Transp. Sci.* 48 (4), 500–520.
- The Encyclopedia of Bus Transport in Hong Kong, 2016. hkbus.wikia.com. Hong Kong Special Administrative Region.
- Thein, S., Chang, Y.S., 2014. Decision making model for lifecycle assessment of lithium-ion battery for electric vehicle—a case study for smart electric bus project in Korea. *J. Power Sources* 249, 142–147.
- Transport Department, 2014. Annual Transport Digest 2014. Hong Kong Special Administrative Region.
- US Environmental Protection Agency, 2008. Toxic air pollutants. <http://www.epa.gov/oar/toxicair/newtoxics.html>.
- Wang, D.Z.W., Lo, H.K., 2008. Multi-fleet ferry service network design with passenger preferences for differential services. *Transp. Res. Part B: Methodol.* 42, 798–822.
- Zhu, Chao, Chen, Xiaohong, 2013. Optimizing battery electric bus transit vehicle scheduling with battery exchanging: model and case study. *Proc.-Social Behav. Sci.* 96, 2725–2736.